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Yuh et al.

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(54) **MRAM CELL AND MRAM**

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G11C 5/02 (2006.01)
G11C 11/16 (2006.01)
H10B 61/00 (2023.01)

(52) **U.S. Cl.**

CPC **H10N 50/80** (2023.02); **G11C 5/025** (2013.01); **G11C 11/161** (2013.01); **H10B 61/00** (2023.02)

(58) **Field of Classification Search**

CPC H10N 50/80; G11C 5/025; G11C 11/161; H10B 61/00
See application file for complete search history.

(56)

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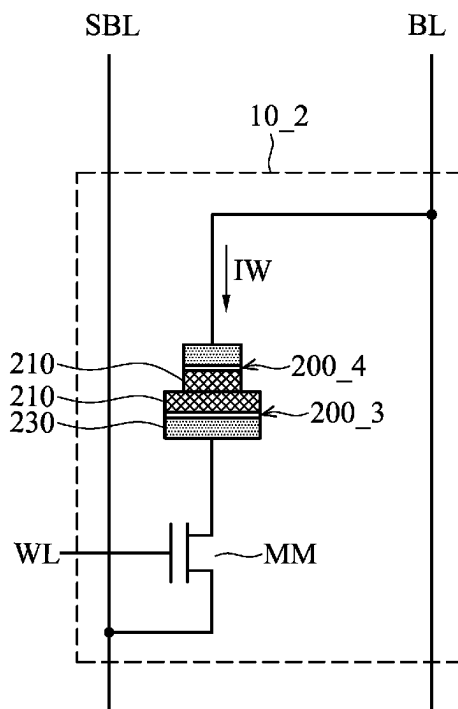
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(57)

ABSTRACT

Magnetic random access memory (MRAM) cells are provided. MRAM cell includes a plurality of stacked magnetic tunnel junction (MTJ) devices coupled in serial and coupled between a bit line and a source line. The stacked MTJ devices have different sizes. Each of the stacked MTJ devices includes a free layer, a pinned layer and a barrier layer between the free layer and the pinned layer. The free layers of two adjacent stacked MTJ devices are in direct contact with each other.

20 Claims, 18 Drawing Sheets



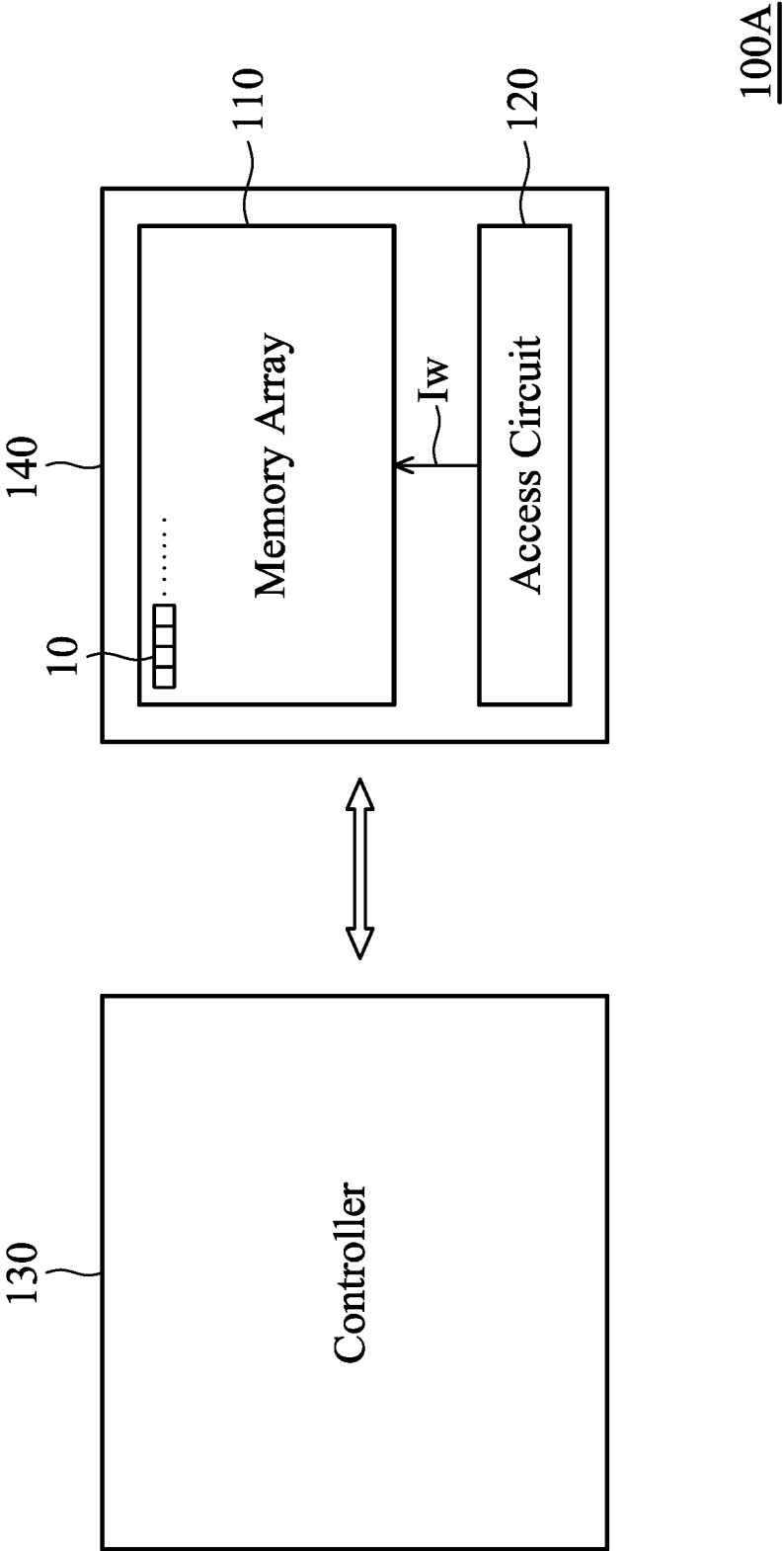


FIG. 1A

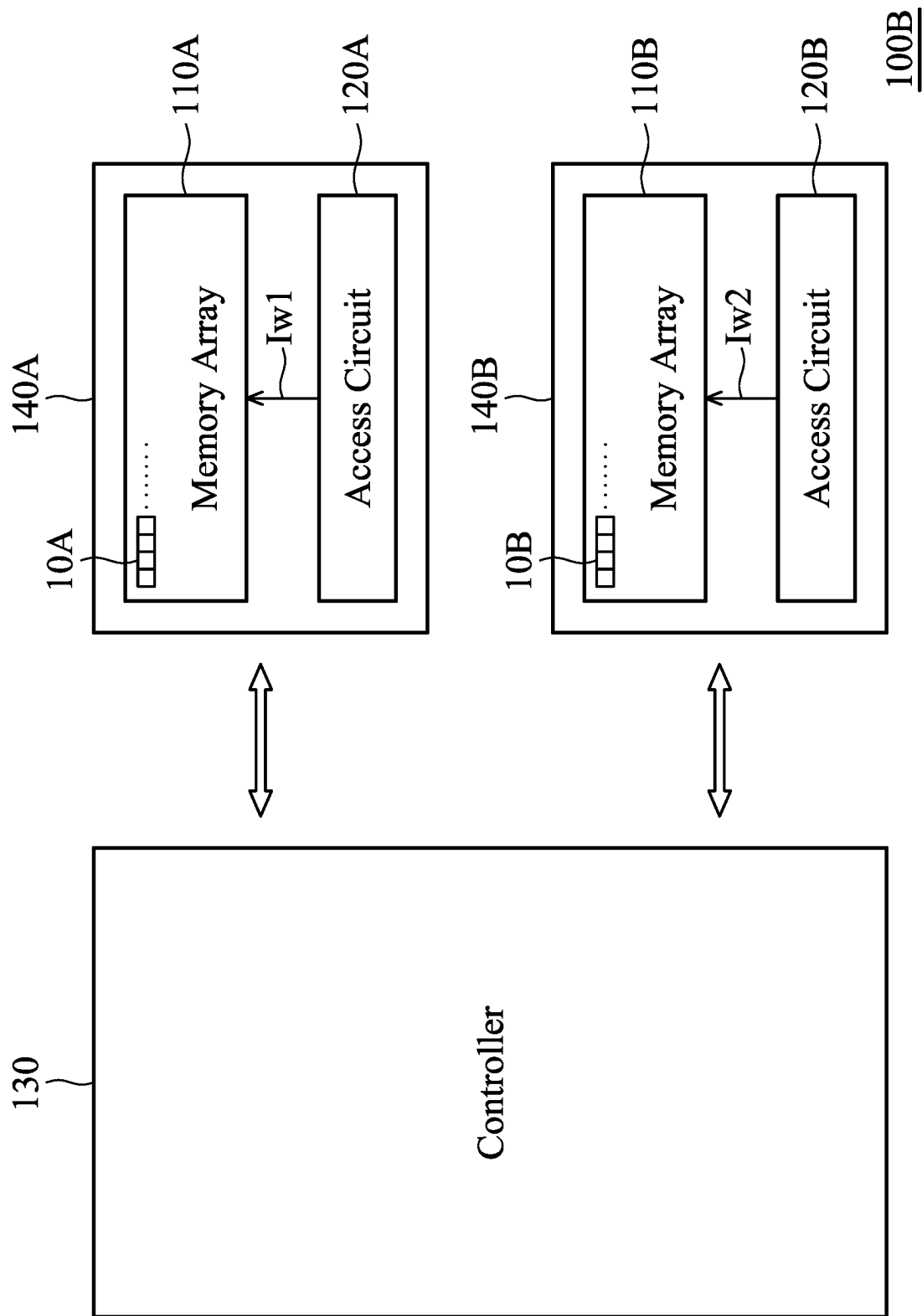


FIG. 1B

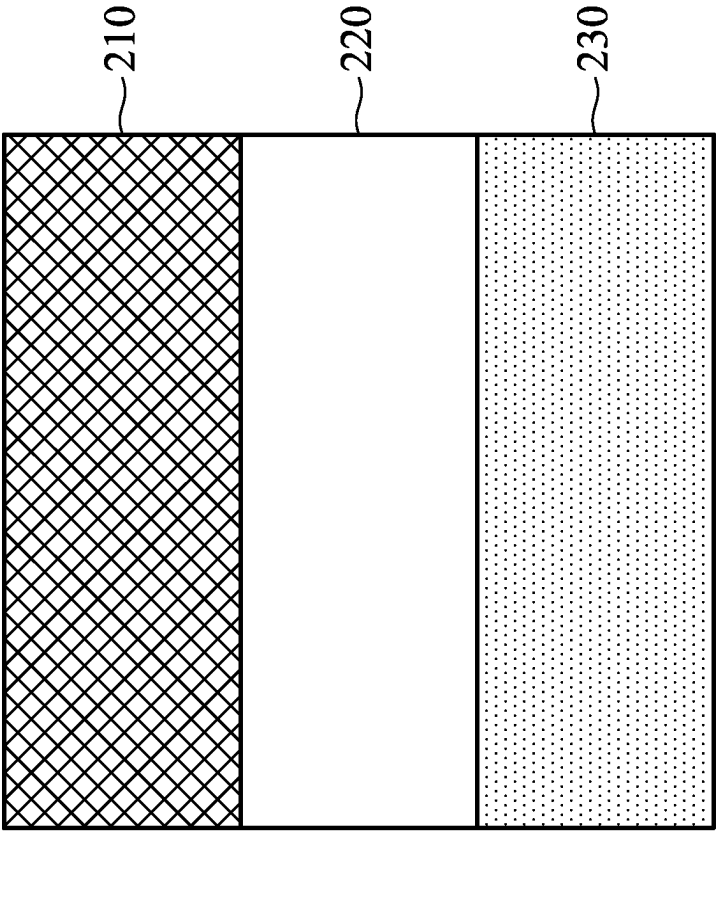


FIG. 2A

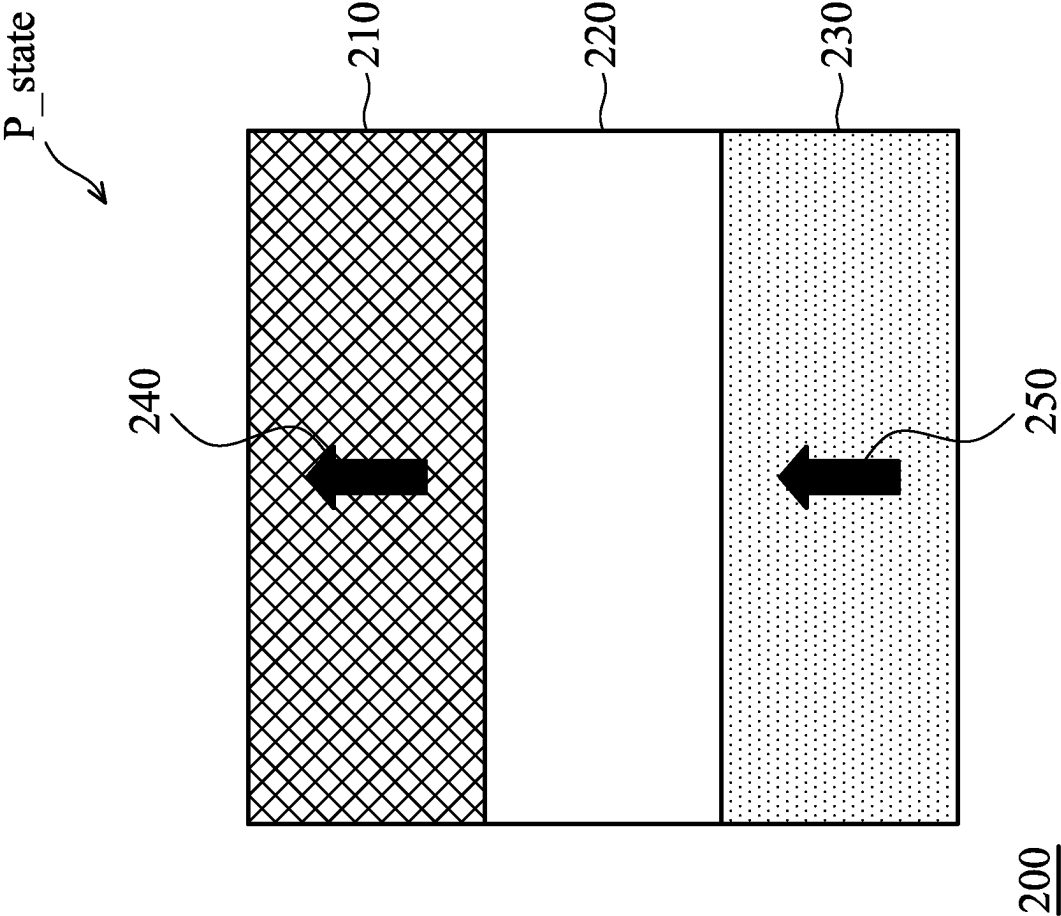


FIG. 2B

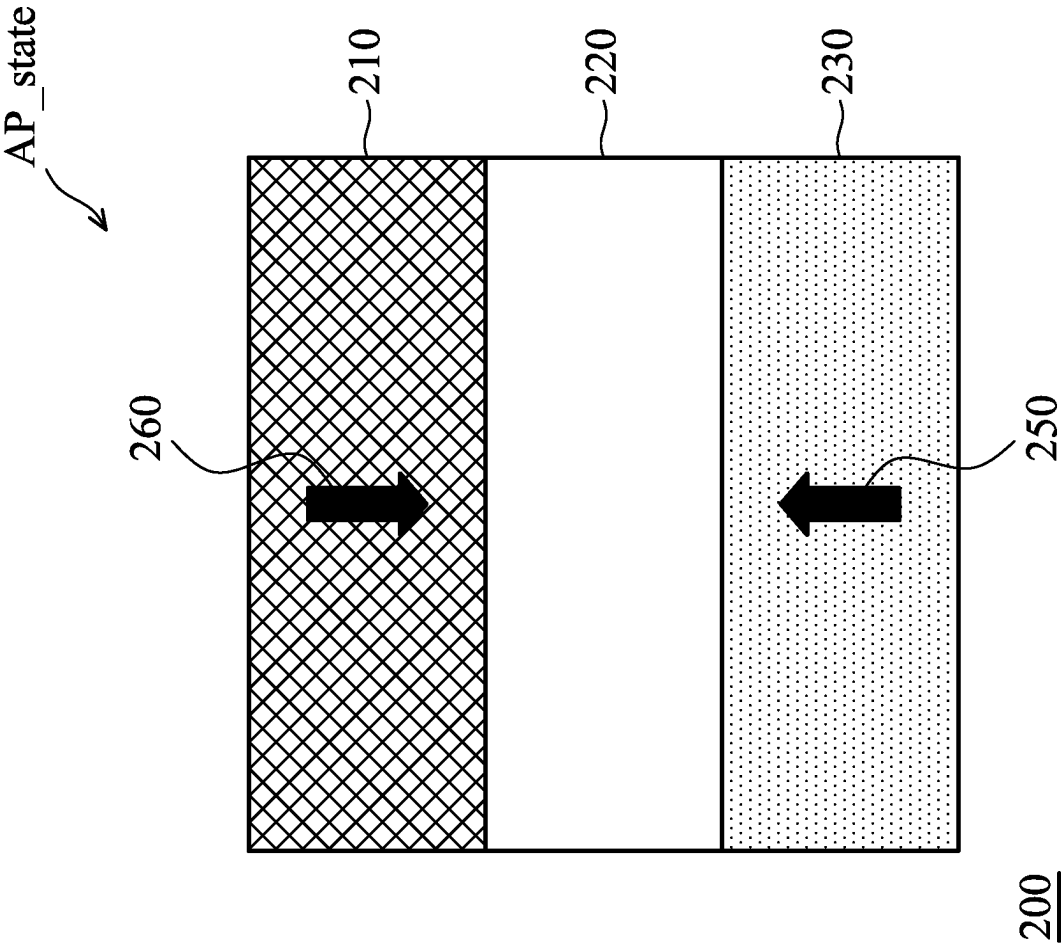


FIG. 2C

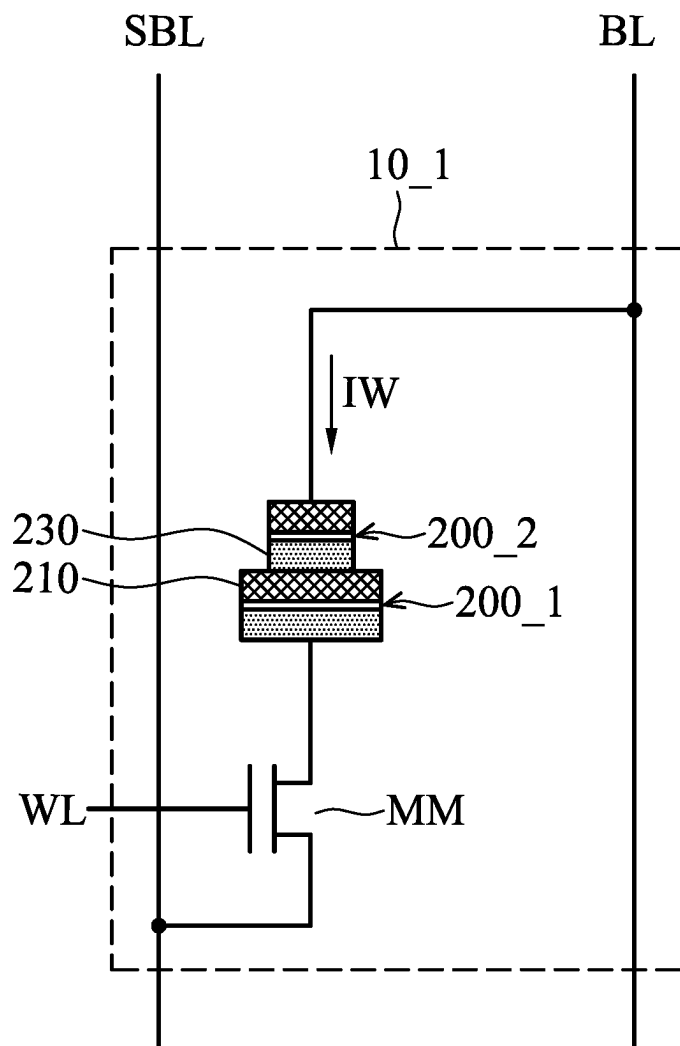


FIG. 3A

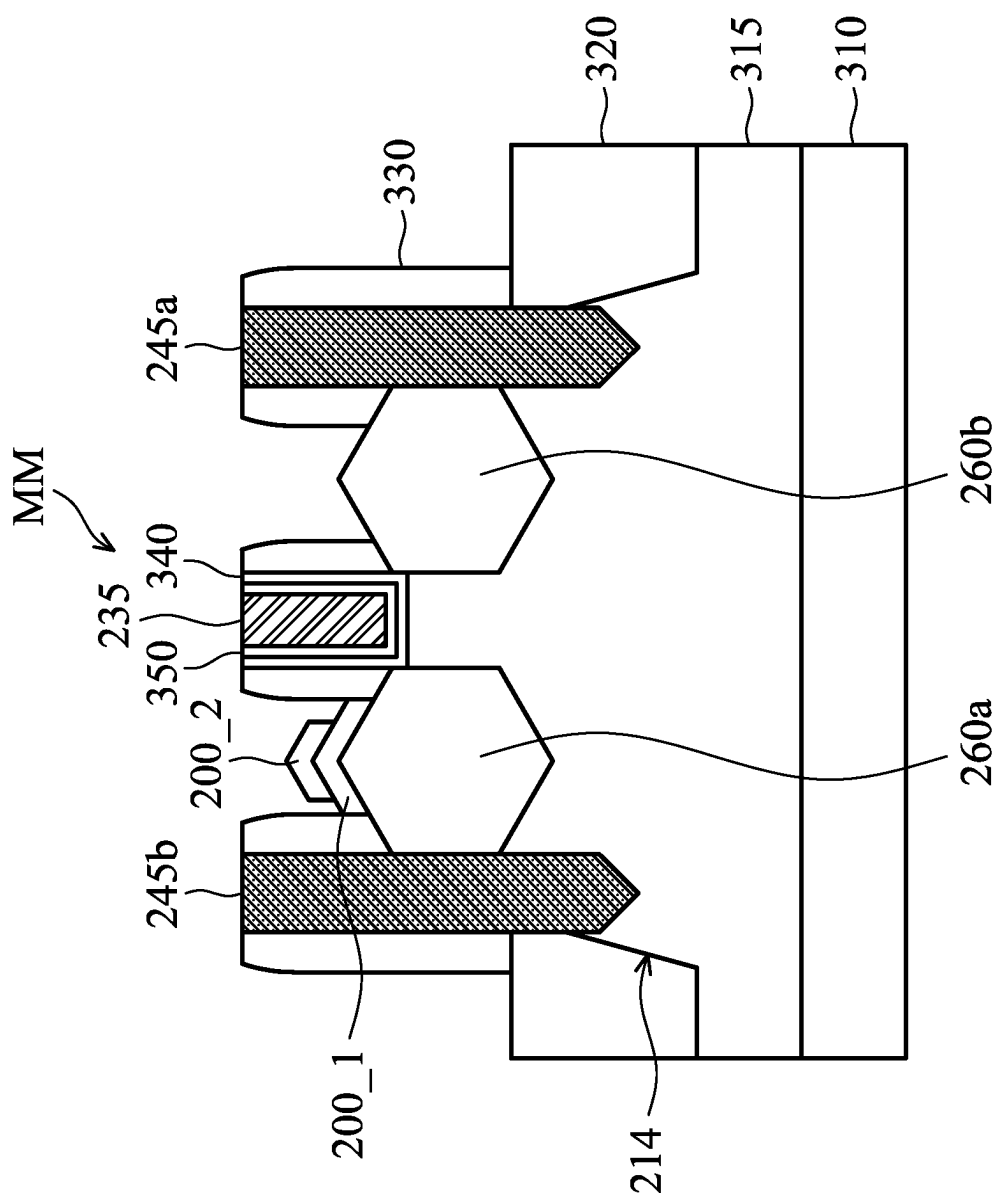


FIG. 3B

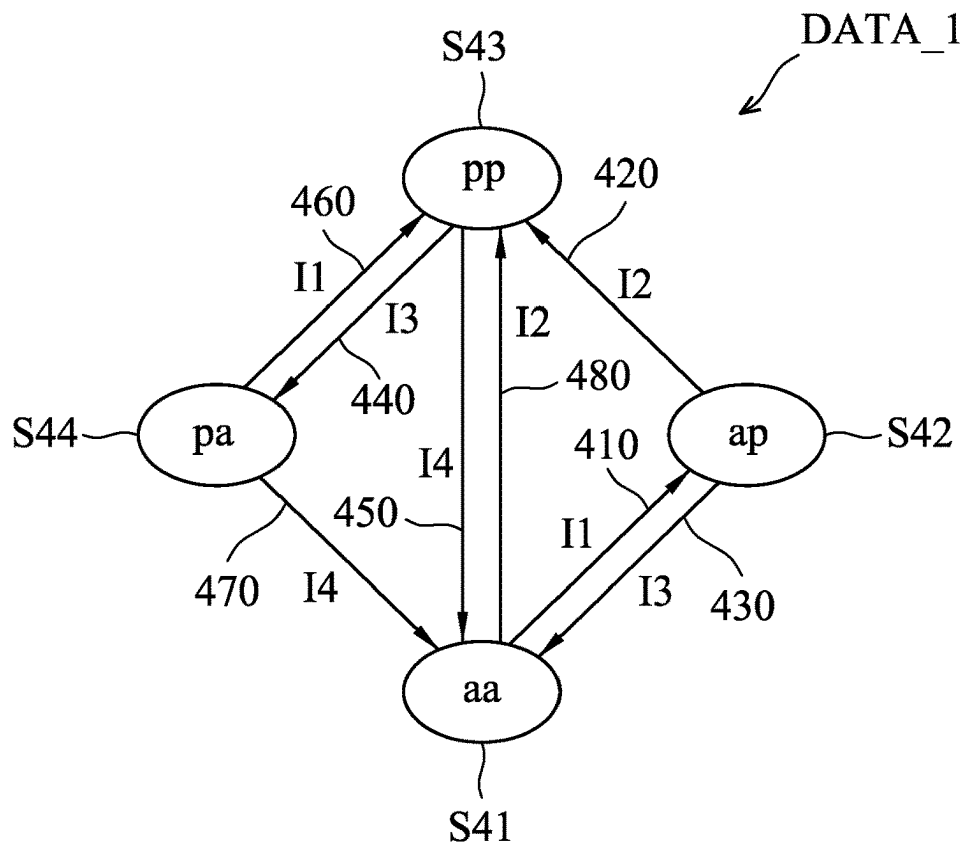


FIG. 4A

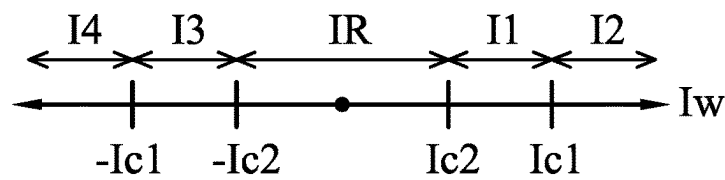


FIG. 4B

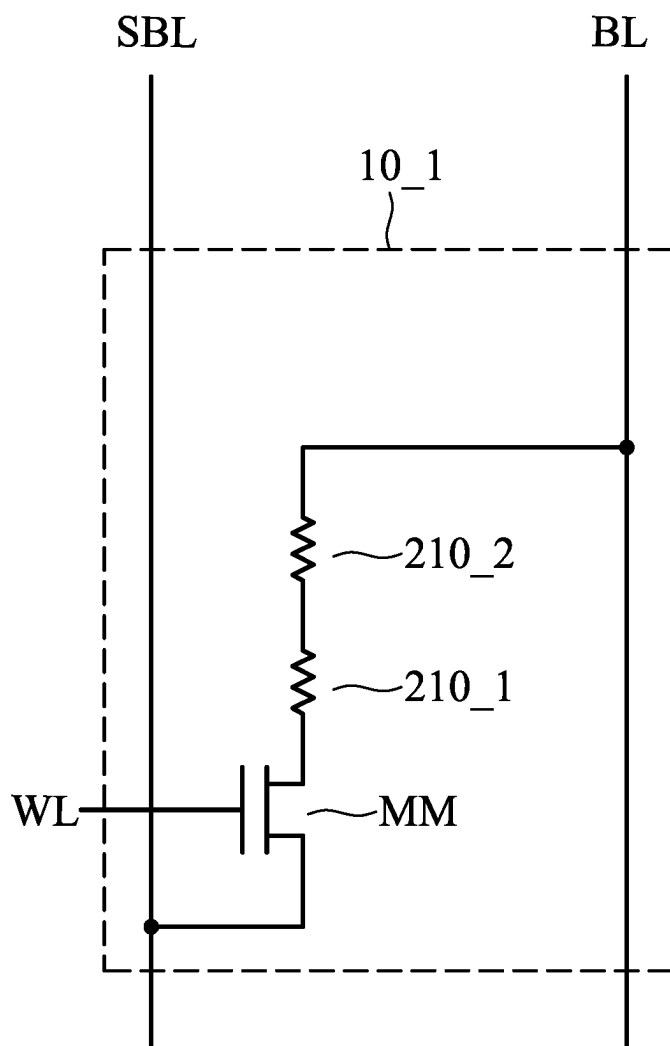


FIG. 5

Access Current I _w	Initial	I1 (Ic2<Iw<Ic1)	I2 (Ic1<Iw)	I3 (-Ic1<Iw<-Ic2)	I4 (Iw<-Ic1)
MTJ1	Rap1	Rap1	Rp1	Rp1	Rap1
MTJ2	Rap2	Rp2	Rp2	Rap2	Rap2
state	aa (S41)	ap (S42)	pp (S43)	pa (S44)	aa (S41)
Serial-R	Rap1+Rap2	Rap1+Rp2	Rp1+Rp2	Rp1+Rap2	Rap1+Rap2

FIG. 6

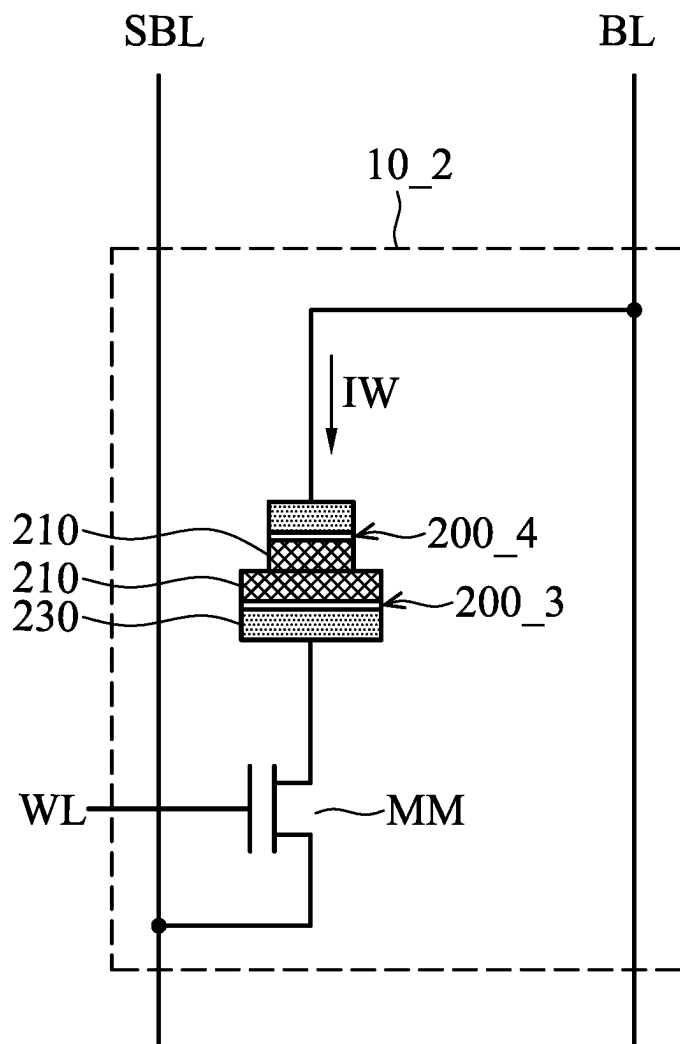


FIG. 7

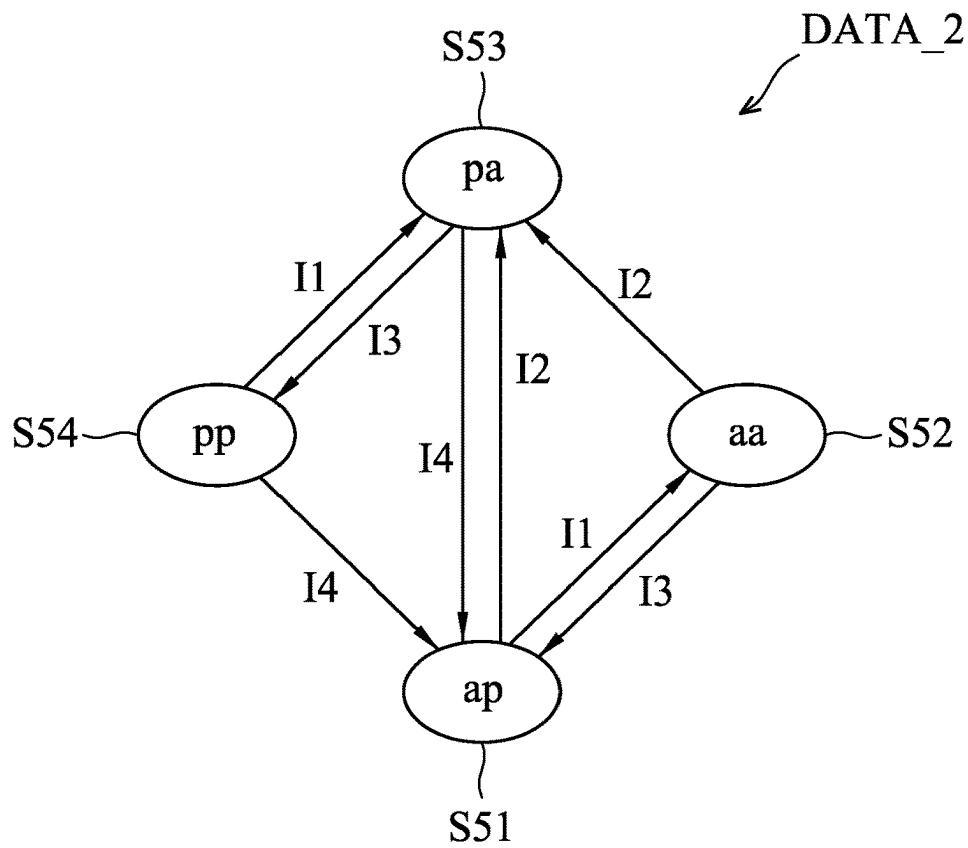


FIG. 8A

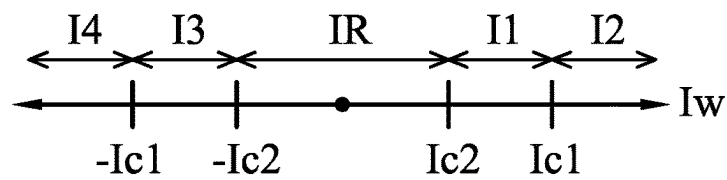


FIG. 8B

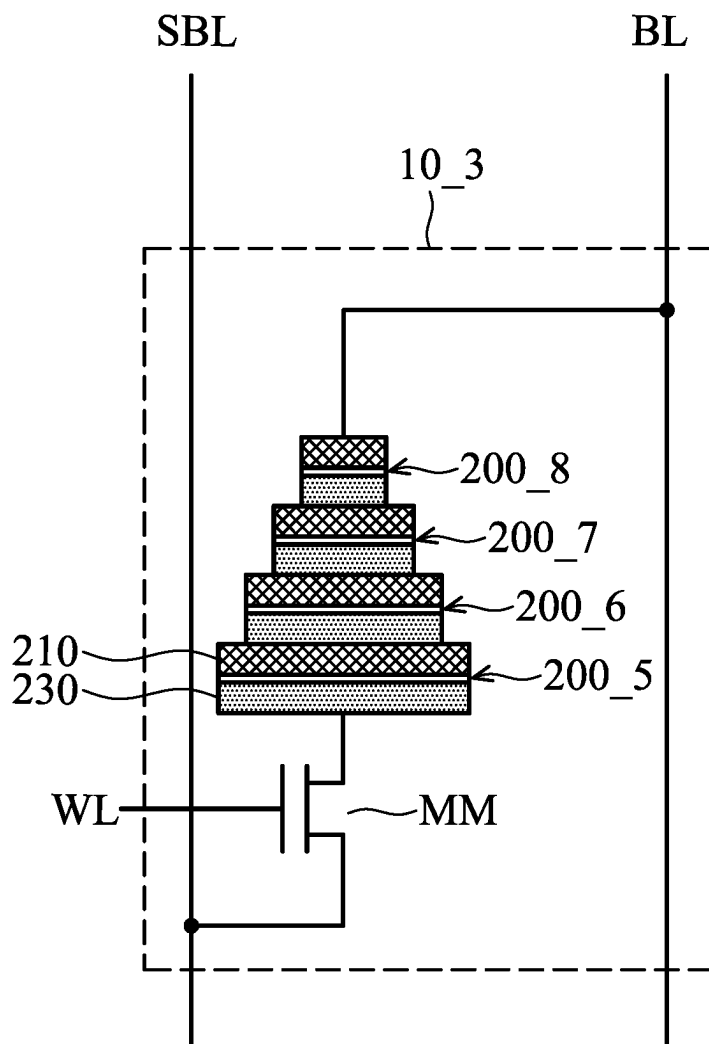


FIG. 9

FIG. 10B

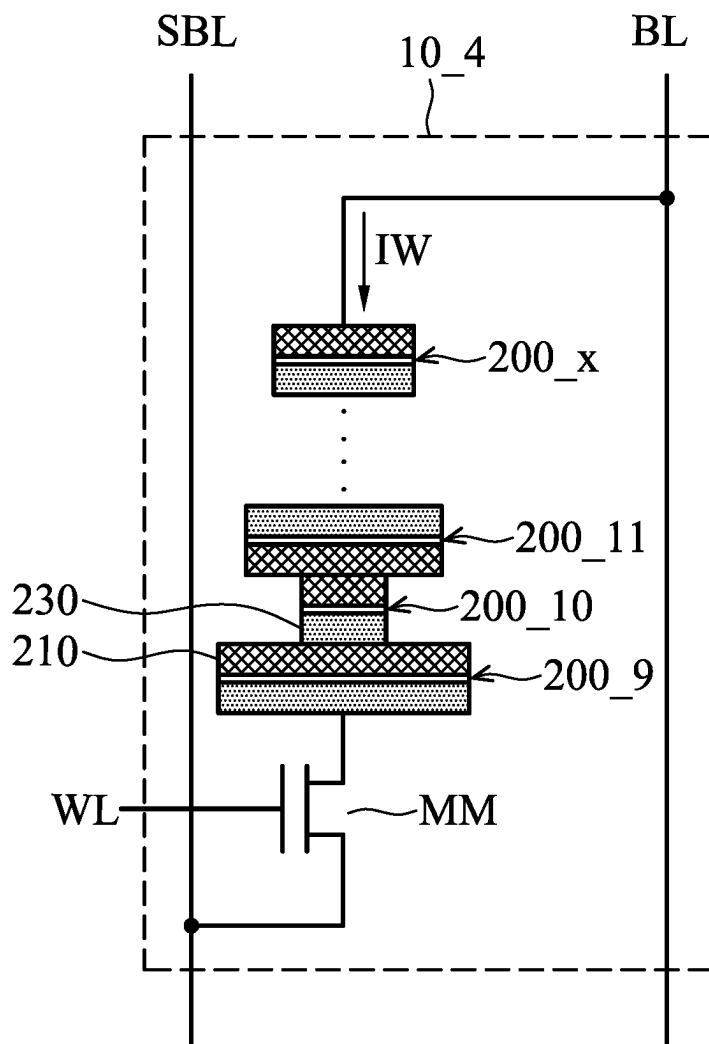


FIG. 11

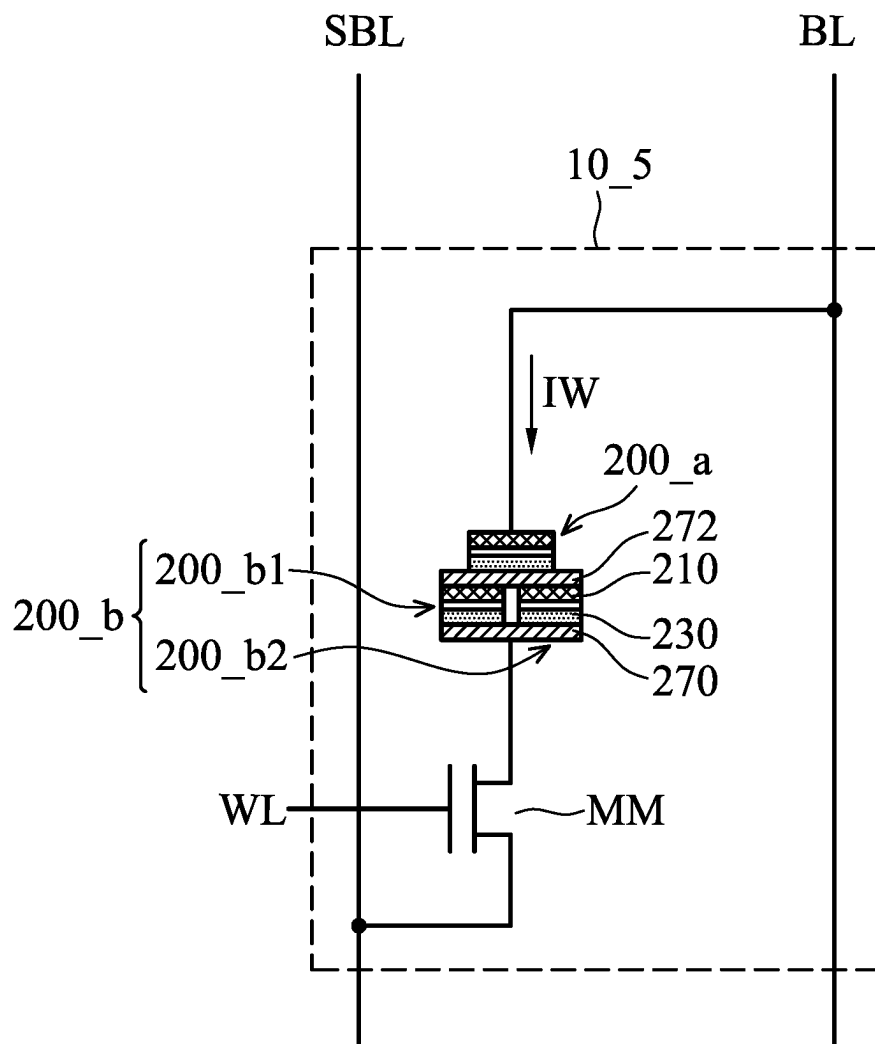


FIG. 12

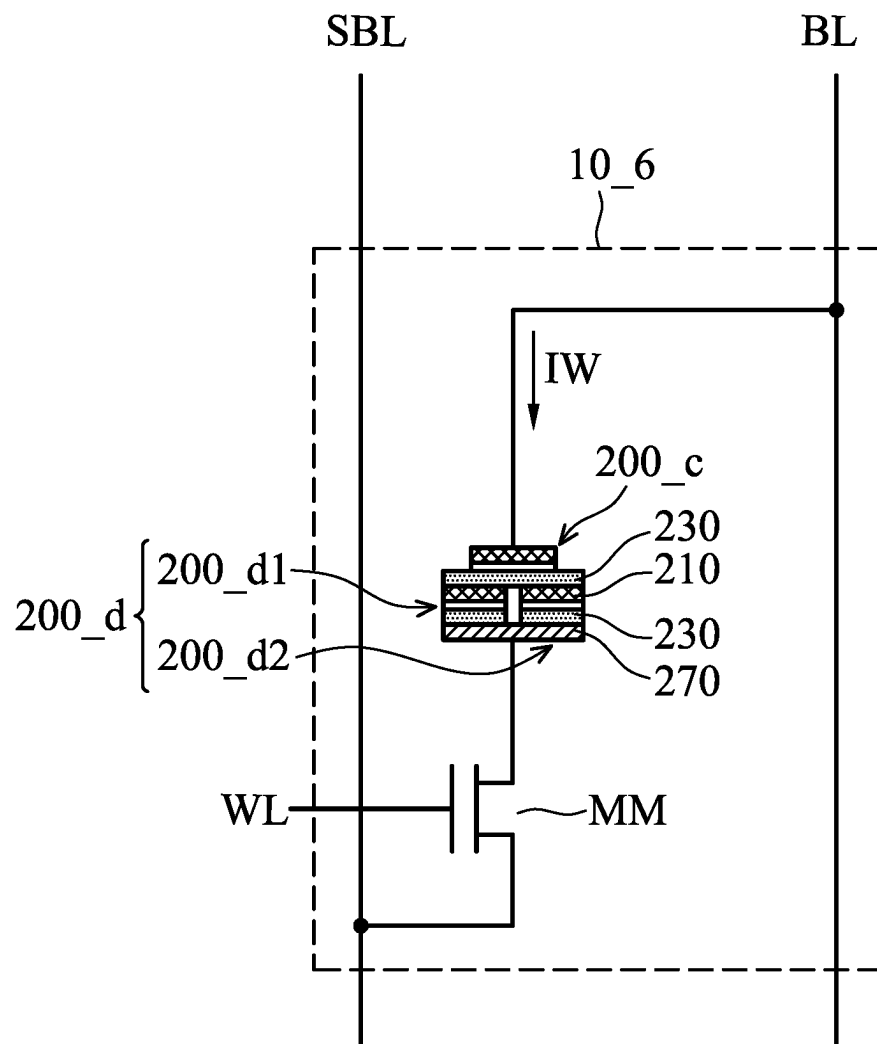


FIG. 13

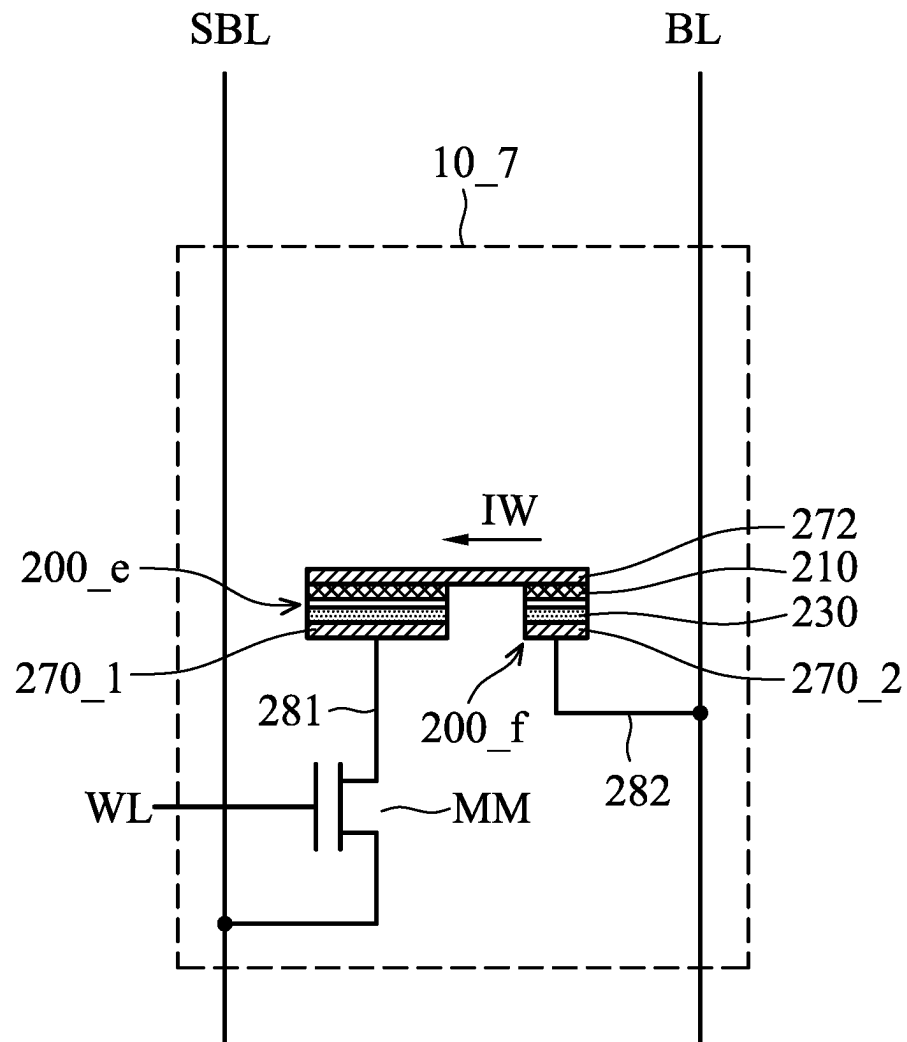


FIG. 14

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MRAM CELL AND MRAM**CROSS REFERENCE TO RELATED APPLICATIONS**

This Application is a Continuation of application Ser. No. 17/734,374, filed on May 2, 2022 now U.S. Pat. No. 11,963,463), which is a Divisional of application Ser. No. 16/884,297, filed on May 27, 2020 (now U.S. Pat. No. 11,322,680), the entirety of which is incorporated by reference herein.

BACKGROUND

An MRAM (magnetic random access memory) is a form of non-volatile memory capable of storing bits of digital information (“0” or “1”). In an MRAM, digital data is not stored as an electrical charge as it is in traditional RAM components, but rather the bit state (“0” or “1”) is stored via resistive states (high resistance or low resistance) in magnetic storage elements which do not require constant electrical power to retain their state.

MRAM may be a cost-effective solution for long-term data retention and rapid on/off applications such as mobile devices and general consumer electronic systems. Compared with static random access memory (SRAM) and flash memory, MRAM can provide fast, low-power, nonvolatile storage.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various nodes are not drawn to scale. In fact, the dimensions of the various nodes may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1A shows an integrated circuit (IC), in accordance with some embodiments of the disclosure.

FIG. 1B shows an IC, in accordance with some embodiments of the disclosure.

FIG. 2A shows an example schematic of an MTJ device, in accordance with some embodiments of the disclosure.

FIG. 2B shows an example of a parallel magnetic state P_{state} of the MTJ device of FIG. 2A.

FIG. 2C shows an example of an anti-parallel magnetic state AP_{state} of the MTJ device of FIG. 2A.

FIG. 3A shows an example schematic of an MRAM cell, in accordance with some embodiments of the disclosure.

FIG. 3B shows a semiconductor structure of the MRAM cell of FIG. 3A, in accordance with some embodiments of the disclosure.

FIG. 4A shows an example state transition diagram of the data $DATA_1$ in the MRAM cell of FIG. 3A, in accordance with some embodiments of the disclosure.

FIG. 4B shows an example illustrating the access current I_w corresponding to the state transition diagram of the data $DATA_1$ in FIG. 4A, in accordance with some embodiments of the disclosure.

FIG. 5 shows an example schematic illustrating an equivalent circuit of the MRAM cell in FIG. 3A, in accordance with some embodiments of the disclosure.

FIG. 6 shows a table illustrating the relationship of the access current I_w and the equivalent impedance of stacked MTJ devices.

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FIG. 7 shows an example schematic of an MRAM cell, in accordance with some embodiments of the disclosure.

FIG. 8A shows an example state transition diagram of the data $DATA_2$ in the MRAM cell of FIG. 7, in accordance with some embodiments of the disclosure.

FIG. 8B shows an example illustrating the access current I_w corresponding to the state transition diagram of the in FIG. 8A, in accordance with some embodiments of the disclosure.

FIG. 9 shows an example schematic of an MRAM cell, in accordance with some embodiments of the disclosure.

FIG. 10A shows an example state transition diagram of the data $DATA_3$ in the MRAM cell of FIG. 9, in accordance with some embodiments of the disclosure.

FIG. 10B shows an example illustrating the access current I_w corresponding to the state transition diagram of the data $DATA_3$ in FIG. 10A, in accordance with some embodiments of the disclosure.

FIG. 11 shows an example schematic of an MRAM cell, in accordance with some embodiments of the disclosure.

FIG. 12 shows an example schematic of an MRAM cell, in accordance with some embodiments of the disclosure.

FIG. 13 shows an example schematic of an MRAM cell, in accordance with some embodiments of the disclosure.

FIG. 14 shows an example schematic of an MRAM cell, in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different nodes of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. In some embodiments, the formation of a first node over or on a second node in the description that follows may include embodiments in which the first and the second nodes are formed in direct contact, and may also include embodiments in which additional nodes may be formed between the first and the second nodes, such that the first and the second nodes may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Some variations of the embodiments are described. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements. It should be understood that additional operations can be provided before, during, and/or after a disclosed method, and some of the operations described can be replaced or eliminated for other embodiments of the method.

Furthermore, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

FIG. 1A shows an integrated circuit (IC) 100A, in accordance with some embodiments of the disclosure. The IC

100A includes a magnetic random access memory (MRAM) 140, and a controller 130. The MRAM 140 includes a memory array 110 and an access circuit 120. The memory array 110 is formed by multiple MRAM cells 10 arranged in multiple rows and multiple columns. The MARM cells 10 of the memory array 110 are used to store data, and each memory cell 10 includes multiple magnetic tunnel junction (MTJ) devices capable of storing multiple-bit data. The access circuit 120 includes circuits for accessing the memory array 110, such as a row decoder, a column decoder, a sense circuit having multiple sense amplifiers, and a writing circuit having multiple writing drivers. According to the address information from the controller 130, the access circuit 120 is capable of providing an access current I_w to perform a read operation to read data from the memory array 110 and/or perform a write operation to write (or program) data into the memory array 110.

FIG. 1B shows an IC 100B, in accordance with some embodiments of the disclosure. The IC 100B includes a first MRAM 140A, a second MRAM 140B and a controller 130. Compared with the IC 100A FIG. 1, the IC 100B includes multiple MRAMs.

The first MRAM 140A includes a memory array 110A and an access circuit 120A. The memory array 110A is formed by multiple MRAM cells 10A arranged in multiple rows and multiple columns. The MARM cells 10A of the memory array 110A are used to store data, and each memory cell 10A includes one or more MTJ device capable of storing at least one bit data. The access circuit 120A includes circuits for accessing the memory array 110A, such as a row decoder, a column decoder, a sense circuit having multiple sense amplifiers, and a writing circuit having multiple writing drivers. According to the address information from the controller 130, the access circuit 120A is capable of providing an access current I_{w1} to perform a read operation to read data from the memory array 110A and/or perform a write operation to write (or program) data into the memory array 110A.

The second MRAM 140B includes a memory array 110B and an access circuit 120B. The memory array 110B is formed by multiple MRAM cells 10B arranged in multiple rows and multiple columns. The MARM cells 10B of the memory array 110B are used to store data, and each memory cell 10B includes multiple MTJ devices capable of storing multiple-bit data. The access circuit 120B includes circuits for accessing the memory array 110B, such as a row decoder, a column decoder, a sense circuit having multiple sense amplifiers, and a writing circuit having multiple writing drivers. According to the address information from the controller 130, the access circuit 120B is capable of providing an access current I_{w2} to perform a read operation to read data from the memory array 110B and/or perform a write operation to write (or program) data into the memory array 110B.

It should be noted that the number of MTJ devices in each memory cell 10A of the first MRAM 140A is different from the number of MTJ devices in each memory cell 10B of the second MRAM 140B. In some embodiments, the number of MTJ devices in each memory cell 10A of the first MRAM 140A is less than the number of MTJ devices in each memory cell 10B of the second MRAM 140B. Thus, the bit number of data stored in the memory cell 10A is less than the bit number of data stored in the memory cell 10B.

FIG. 2A shows an example schematic of an MTJ device 200, in accordance with some embodiments of the disclosure. The MTJ device 200 includes at least the following layers: a free layer 210, a barrier layer 220 which serves as a tunneling barrier, and a pinned layer 230. In each MTJ

device 200, the pinned layer 230 has a fixed magnetization direction (or magnetic orientation). Furthermore, the free layer 210 has a magnetization direction that, with the application of a spin torque caused by a current through the MTJ device 200, can be switched between a first direction, which is parallel to the magnetization direction of the pinned layer 230 (i.e. a parallel magnetic state P_state), and a second direction opposite to the first direction. The second direction is anti-parallel to the magnetization direction of the pinned layer 230 (i.e. an anti-parallel magnetic state AP_state). In some embodiments, the free layer 210, the barrier layer 220 and the pinned layer 230 are formed in the back end of line (BEOL) process.

The barrier layer 220 is disposed between the pinned layer 230 and the free layer 210. The barrier layer 220 is composed of insulating material, for example aluminum oxide, magnesium oxide, or tantalum oxide. The barrier layer 220 is formed thin enough to allow the transfer (tunneling) of spin-aligned electrons when the magnetization directions of the free layer 210 and the pinned layer 230 are parallel. Conversely, when the magnetization directions of the free layer 210 and the pinned layer 230 are anti-parallel, the probability of electrons tunneling through the barrier layer 220 is reduced.

FIG. 2B shows an example of a parallel magnetic state P_state of the MTJ device 200 of FIG. 2A. In FIG. 2B, the free layer 210 has a magnetization direction 240 from the pinned layer 230 to the free layer 210. Furthermore, the pinned layer 230 has a magnetization direction 250 from the pinned layer 230 to the free layer 210. Due to the magnetization direction 240 being parallel to the magnetization direction 250, the MTJ device 200 is in a parallel magnetic state P_state with a low resistance R_p . The resistance of the MTJ device 200 is decreased when the magnetization directions 240 and 250 become more parallel.

FIG. 2C shows an example of an anti-parallel magnetic state AP_state of the MTJ device 200 of FIG. 2A. In FIG. 2C, the free layer 210 has a magnetization direction 260 from the free layer 210 to the pinned layer 230. Furthermore, the pinned layer 230 has a magnetization direction 250 from the pinned layer 230 to the free layer 210. Due to the magnetization direction 260 being anti-parallel to the magnetization direction 250, the MTJ device 200 is in an anti-parallel magnetic state AP_state with a high resistance R_{ap} higher than the low resistance R_p , i.e., $R_{ap} > R_p$. The resistance of the MTJ device 200 is increased when the magnetization directions 260 and 250 become more anti-parallel.

In such embodiments, the MTJ device 200 is a perpendicular-spin MTJ, and the spins of the free layer 210 and the pinned layer 230 are perpendicular to the plane of the layers themselves.

In some embodiments, the MTJ device 200 is an in-plane-spin MTJ, and the spins of the free layer 210 and the pinned layer 230 are in the same plane as the layers themselves.

In the MRAM, the resistance of the MTJ device 200 can be switched between the high resistance R_{ap} and the low resistance R_p that respectively represent first and second logic levels. For example, the high resistance R_{ap} of an anti-parallel magnetic state AP_state can represent a high logic level (i.e. "1"), and the low resistance R_p of a parallel magnetic state P_state can represent a low logic state (i.e. "0"). Thus, by controlling the magnetization directions of the free layer 210 and the pinned layer 230 of the MTJ device 200 with a writing current, a bit ("1" or "0") of data is stored in the MTJ device 200. Furthermore, the logic level of data stored in the MTJ device 200 can be read by passing

a sense current through the MTJ device **200** and then sensing the resistance of the MTJ device **200**. In some embodiments, the writing current and the sense current are provided by the access current (e.g., I_w of FIG. 1A and I_{w1} and I_{w2} of FIG. 1B).

FIG. 3A shows an example schematic of an MRAM cell **10_1**, in accordance with some embodiments of the disclosure. The MRAM cell **10_1** includes a transistor MM and the MTJ devices **200_1** and **200_2** coupled in series. In some embodiments, the transistor MM is a N-type transistor. The transistor MM is coupled between a source line SBL and the MTJ device **200_1**, and a gate of the transistor MM is coupled to a word line WL. The MTJ devices **200_1** and **200_2** are stacked between the transistor MM and a bit line BL. The MRAM cell **10_1** is capable of storing two-bit data DATA_1 based on the stacked MTJ devices **200_1** and **200_2**. As described above, by applying the access current I_w to the stacked MTJ devices **200_1** and **200_2**, the MTJ device **200_1** and/or **200_2** is switched from a parallel magnetic state P_state to an anti-parallel magnetic state AP_state or from the anti-parallel magnetic state AP_state to the parallel magnetic state P_state.

In the MRAM cell **10_1**, the free layer **210** of the MTJ device **200_1** is in contact with the pinned layer **230** of the MTJ device **200_2**. In some embodiments, a conducting layer (e.g., an electrode) is formed between the free layer **210** of the MTJ device **200_1** and the pinned layer **230** of the MTJ device **200_2**. It should be noted that the MTJ devices **200_1** and **200_2** are different sizes.

In some embodiments, the thickness of the free layers **210** of the MTJ devices **200_1** and **200_2** are the same. Furthermore, the thickness of the barrier layer **220** of the MTJ devices **200_1** and **200_2** are the same, and the thickness of the pinned layer **230** of the MTJ devices **200_1** and **200_2** are the same. The difference between the MTJ devices **200_1** and **200_2** is that the area of the free layer **210**, the barrier layer **220** and the pinned layer **230** of the MTJ device **200_1** is greater than that of the MTJ device **200_2**. Thus, the equivalent impedances of the MTJ devices **200_1** and **200_2** are different.

In some embodiments, the thickness of the free layers **210** of the MTJ devices **200_1** and **200_2** are the same, and the thickness of the pinned layer **230** of the MTJ devices **200_1** and **200_2** are the same. Furthermore, the area of the free layer **210**, the barrier layer **220** and the pinned layer **230** of the MTJ device **200_1** is the same as that of the MTJ device **200_2**. The difference between the MTJ devices **200_1** and **200_2** is that the thickness of the barrier layer **220** of the MTJ devices **200_1** and **200_2** are different. Thus, the equivalent impedances of the MTJ devices **200_1** and **200_2** are different.

In some embodiments, the transistor MM is a planar transistor, e.g., CMOS transistor. The MTJ devices **200_1** and **200_2** are planar devices. The stacked MTJ devices **200_1** and **200_2** formed a conventional planar structure that is coupled to the source/drain region of the transistor MM through various interconnection layers.

In some embodiments, the transistor MM is a fin field-effect transistor (FINFET). The stacked MTJ devices **200_1** and **200_2** form a conventional planar structure that is coupled to the source/drain region of the transistor MM through various interconnection layers.

In some embodiments, the transistor MM is a FINFET. The stacked MTJ devices **200_1** and **200_2** form a non-planar structure that is over the source/drain region of the transistor MM without interconnection layers.

FIG. 3B shows a semiconductor structure of the MRAM cell **10_1** of FIG. 3A, in accordance with some embodiments of the disclosure. A well region **315** is formed over a semiconductor substrate **310**. In some embodiments, the semiconductor substrate **310** is a Si substrate. In some embodiments, the material of the semiconductor substrate **310** is selected from a group consisting of bulk-Si, SiP, SiGe, SiC, SiPC, Ge, SOI-Si, SOI-SiGe, III-VI material, and a combination thereof. A semiconductor fin **214** is formed on the well region **315**, and is surrounded by the shallow trench isolation (STI) **320**.

The source/drain regions **260a** and **260b** of the transistor MM are formed on the semiconductor fin **214**. In some embodiments, the source/drain silicide regions (not shown) are formed on the source/drain regions **260a** and **260b**. The gate electrode **235** is formed over the gate dielectric layer **340** and is positioned over a top surface of the semiconductor fin **214**. Thus, the gate electrode **235** and the gate dielectric layer **340** over the semiconductor fin **214** form a gate structure for the transistor MM. A work-function layer **350** is formed between the gate dielectric layer **340** and the gate electrode **235**. In some embodiments, the spacers **330** are formed on opposite sides of the gate electrode **235**. The spacers **330** may be formed by a single layer or multiple layers.

For the transistor MM, the channel region corresponding to the semiconductor fin **214** is formed between the source/drain regions **260a** and **260b** and under the gate electrode **235**. Furthermore, the dielectric-base gates **245a** and **245b** are located upon the edge of the semiconductor fin **214**. For example, the dielectric-base gate **245b** is arranged upon the left edge of the semiconductor fin **214**, and the dielectric-base gate **245a** is arranged upon the right edge of the semiconductor fin **214**. Furthermore, the dielectric-base gates **245a** and **245b** are deeper than the source/drain regions **260a** and **260b**. A contact (not shown) is formed on the source/drain region **260b** for connecting the source line SBL through various interconnection layers (not shown), such as the via layers and the metal layers.

In FIG. 3B, the MTJ devices **200_1** and **200_2** are formed over the source/drain region **260a**. In some embodiments, the MTJ device **200_1** is formed on the source/drain region **260a** and along the surface of the source/drain region **260a**, and the MTJ device **200_2** is formed on the MTJ device **200_1** and along the surface of the MTJ device **200_1**. Therefore, the MTJ devices **200_1** and **200_2** are non-planar. Furthermore, the MTJ device **200_2** has smaller size, and cannot completely cover the MTJ device **200_1**. In some embodiments, the MTJ device **200_1** is in contact with the MTJ device **200_2** and the source/drain region **260a**.

FIG. 4A shows an example state transition diagram of the data DATA_1 in the MRAM cell **10_1** of FIG. 3A, in accordance with some embodiments of the disclosure. FIG. 4B shows an example illustrating the access current I_w corresponding to the state transition diagram of the data DATA_1 in FIG. 4A, in accordance with some embodiments of the disclosure.

In some embodiments, the MTJ devices **200_1** and **200_2** are initially operated in the anti-parallel magnetic state AP_state, the data DATA_1 has a first state S41, i.e., "aa". The first state "a" represents that the MTJ device **200_1** is operated in the anti-parallel magnetic state AP_state, and the second state "a" represents that the MTJ device **200_2** is operated in the anti-parallel magnetic state AP_state. As described above, the high resistance R_{ap} of the anti-parallel magnetic state AP_state can represent a high logic level (i.e. "1"). The MTJ device **200_2** has smaller size than the MTJ

device **200_1**. Therefore, for the same access current I_w , the MTJ device **200_2** has higher current density.

If the access current I_w within a current region **I1** is applying into the stacked MTJ devices **200_1** and **200_2** of the MRAM cell **10_1**, the MTJ device **200_2** is switched from the anti-parallel magnetic state AP_state to the parallel magnetic state P_state due to higher current density, as shown in label **410**. Thus, the data DATA_1 has a second state **S42**, i.e., “ap”. The state “p” represents that the MTJ device **200_2** is operated in the parallel magnetic state P_state. As described above, the low resistance R_p of the parallel magnetic state P_state can represent a low logic level (i.e. “0”). The current region **I1** is between the current values I_{c2} and I_{c1} . In other words, for the data DATA_1, the access current I_w that is capable of switching the first state **S41** to the second state **S42**, is less than the current value I_{c1} and greater than the current value I_{c2} .

In the second state **S42**, if the access current I_w within a current region **12** is applying into the stacked MTJ devices **200_1** and **200_2** of the MRAM cell **10_1**, the MTJ device **200_1** is switched from the anti-parallel magnetic state AP_state to the parallel magnetic state P_state, as shown in label **420**. Thus, the data DATA_1 has a third state **S43**, i.e., “pp”. The current region **12** is greater than the current value I_{c1} . In other words, for the data DATA_1 with the second state **S42**, the access current I_w that is capable of switching the second state **S42** to the third state **S43**, is greater than the current value I_{c1} .

In the second state **S42**, if the access current I_w within a current region **13** is applying into the stacked MTJ devices **200_1** and **200_2** of the MRAM cell **10_1**, the MTJ device **200_2** is switched from the parallel magnetic state P_state to the anti-parallel magnetic state AP_state, as shown in label **430**. Thus, the data DATA_1 has the first state **S41**. The current region **13** is between the current values $-I_{c2}$ and $-I_{c1}$. In other words, for the data DATA_1 with the second state **S42**, the access current I_w that is capable of switching the second state **S42** to the first state **S41**, is greater than the current value $-I_{c1}$ and less than the current value $-I_{c2}$.

In the third state **S43**, if the access current I_w within a current region **14** is applying into the stacked MTJ devices **200_1** and **200_2** of the MRAM cell **10_1**, the MTJ devices **200_1** and **200_2** are switched from the parallel magnetic state P_state to the anti-parallel magnetic state AP_state, as shown in label **450**. Thus, the data DATA_1 has the first state **S41**. The current region **14** is less than the current value $-I_{c1}$. In other words, for the data DATA_1 with the third state **S43**, the access current I_w that is capable of switching the third state **S43** to the first state **S41**, is less than the current value $-I_{c1}$.

In the third state **S43**, if the access current I_w within the current region **13** is applying into the stacked MTJ devices **200_1** and **200_2** of the MRAM cell **10_1**, the MTJ device **200_2** is switched from the parallel magnetic state P_state to the anti-parallel magnetic state AP_state, as shown in label **440**. Thus, the data DATA_1 has a fourth state **S44**, i.e., “pa”.

In the fourth state **S44**, if the access current I_w within the current region **I1** is applying into the stacked MTJ devices **200_1** and **200_2** of the MRAM cell **10_1**, the MTJ device **200_2** is switched from the anti-parallel magnetic state AP_state to the parallel magnetic state P_state, as shown in label **470**. Thus, the data DATA_1 has the third state **S43**.

In the fourth state **S44**, if the access current I_w within the current region **14** is applying into the stacked MTJ devices **200_1** and **200_2** of the MRAM cell **10_1**, the MTJ device **200_1** is switched from the parallel magnetic state P_state to

the anti-parallel magnetic state AP_state, as shown in label **470**. Thus, the data DATA_1 has the first state **S41**.

In the first state **S41**, if the access current I_w within the current region **12** is applying into the stacked MTJ devices **200_1** and **200_2** of the MRAM cell **10_1**, the MTJ devices **200_1** and **200_2** are switched from the anti-parallel magnetic state AP_state to the parallel magnetic state P_state, as shown in label **480**. Thus, the data DATA_1 has the third state **S43**.

Specifically, the access current I_w within the current regions **I1** through **14** is a writing current for storing the two-bit data DATA_1 in the MRAM cell **10_1**. Conversely, the access current I_w within the current region **IR** (i.e. $-I_{c2} < I_w < I_{c2}$) is a sense current for reading the two-bit data DATA_1 stored in the MRAM cell **10_1**. In some embodiments, the four states **S41** through **S44** of the stacked MTJ devices **200_1** and **200_2** are used to represent the four logic levels “11”, “10”, “00” and “01” of the two-bit data DATA_1, respectively.

In some embodiments, the access current I_w within the current region **I1** is capable of controlling the magnetization directions of the free layer **210** and the pinned layer **230** of the MTJ device **200_2** as the parallel magnetic state P_state, and the access current I_w within the current region **13** is capable of controlling the magnetization directions of the free layer **210** and the pinned layer **230** of the MTJ device **200_2** as the anti-parallel magnetic state AP_state. The access current I_w within the current region **12** is capable of controlling the magnetization directions of the free layer **210** and the pinned layer **230** of the MTJ device **200_1** as the parallel magnetic state P_state, and the access current I_w within the current region **14** is capable of controlling the magnetization directions of the free layer **210** and the pinned layer **230** of the MTJ device **200_1** as the anti-parallel magnetic state AP_state.

FIG. 5 shows an example schematic illustrating an equivalent circuit of the MRAM cell **10_1** in FIG. 3A, in accordance with some embodiments of the disclosure. In FIG. 5, the resistor **210_1** represents the equivalent resistor of the MTJ device **200_1** of FIG. 3A. Furthermore, the resistor **210_2** represents the equivalent resistor of the MTJ device **200_2** of FIG. 3A. The resistors **210_1** and **210_2** are coupled in series between the transistor MM and the bit line BL. When the MTJ device **200_1** is in the anti-parallel magnetic state AP_state, the equivalent impedance of the resistor **210_1** is R_{ap1} . Conversely, when the MTJ device **200_1** is in the parallel magnetic state P_state, the equivalent impedance of the resistor **210_1** is R_{p1} , where $R_{p1} < R_{ap1}$. Similarly, when the MTJ device **200_2** is in the anti-parallel magnetic state AP_state, the equivalent impedance of the resistor **210_2** is R_{ap2} . Conversely, when the MTJ device **200_2** is in the parallel magnetic state P_state, the equivalent impedance of the resistor **210_2** is R_{p2} , where $R_{p2} < R_{ap2}$.

In the MRAM cell **10_1**, the MTJ device **200_2** has smaller size than the MTJ device **200_1**. For the same access current I_w , the MTJ device **200_2** has larger current density than the MTJ device **200_1**. Thus, the equivalent impedance of the resistor **210_2** corresponding to the MTJ device **200_2** is greater than the equivalent impedance of the resistor **210_1** corresponding to the MTJ device **200_1**, i.e., $R_{p2} > R_{p1}$ and $R_{ap2} > R_{ap1}$.

FIG. 6 shows a table illustrating the relationship of the access current I_w and the equivalent impedance of stacked MTJ devices **200_1** and **200_2**. In FIG. 6, the second row of the table represents the equivalent impedance “MTJ1” of the resistor **210_1** corresponding to the MTJ device **200_1** in response to the access current I_w . The third row of the table

represents the equivalent impedance “MTJ2” of the resistor 210_2 corresponding to the MTJ device 200_2 in response to the access current I_w . The fourth row of the table represents the magnetic states of stacked MTJ devices 200_1 and 200_2 (i.e., the state S41, S42, S43 or S44 of the MRAM cell 10_1) in response to the access current I_w . The fifth row of the table represents the equivalent impedance Serial-R of the stacked MTJ devices 200_1 and 200_2 in response to the access current I_w .

In the second column of the table of FIG. 6, the access current I_w is within the current region IR of FIG. 4B, e.g., an initial current region or a read current region. At this time, the equivalent impedance MTJ1 is R_{p1} , and the equivalent impedance MTJ2 is R_{p2} . Furthermore, the MRAM cell 10_1 has the first state S41 (e.g., “aa”), and the equivalent impedance Serial-R of the stacked MTJ devices 200_1 and 200_2 is equal to $R_{p1}+R_{p2}$.

In the third column of the table of FIG. 6, the access current I_w is within the current region II of FIG. 4B, e.g., $I_{c2} < I_w < I_{c1}$. At this time, the equivalent impedance MTJ1 is R_{p1} , and the equivalent impedance MTJ2 is R_{p2} . Furthermore, the MRAM cell 10_1 has the first state S42 (e.g., “ap”), and the equivalent impedance Serial-R of the stacked MTJ devices 200_1 and 200_2 is equal to $R_{p1}+R_{p2}$.

In the fourth column of the table of FIG. 6, the access current I_w is within the current region 12 of FIG. 4B, e.g., $I_w < I_{c1}$. At this time, the equivalent impedance MTJ1 is R_{p1} , and the equivalent impedance MTJ2 is R_{p2} . Furthermore, the MRAM cell 10_1 has the first state S43 (e.g., “pp”), and the equivalent impedance Serial-R of the stacked MTJ devices 200_1 and 200_2 is equal to $R_{p1}+R_{p2}$.

In the fifth column of the table of FIG. 6, the access current I_w is within the current region 13 of FIG. 4B, e.g., $-I_{c1} < I_w < -I_{c2}$. At this time, the equivalent impedance MTJ1 is R_{p1} , and the equivalent impedance MTJ2 is R_{p2} . Furthermore, the MRAM cell 10_1 has the first state S44 (e.g., “pa”), and the equivalent impedance Serial-R of the stacked MTJ devices 200_1 and 200_2 is equal to $R_{p1}+R_{p2}$.

In the sixth column of the table of FIG. 6, the access current I_w is within the current region 14 of FIG. 4B, e.g., $I_w < -I_{c1}$. At this time, the equivalent impedance MTJ1 is R_{p1} , and the equivalent impedance MTJ2 is R_{p2} . Furthermore, the MRAM cell 10_1 has the first state S41, and the equivalent impedance Serial-R of the stacked MTJ devices 200_1 and 200_2 is $R_{p1}+R_{p2}$.

In FIG. 6, each of the states S41 through S44 has individual equivalent impedance. Thus, by providing the sense signal (current or voltage) to obtain the equivalent impedance Serial-R of the stacked MTJ devices 200_1 and 200_2 in the MRAM cell 10_1, the current state of the MRAM cell 10_1 is obtained. According to the current state of the MRAM cell 10_1, the data DATA_1 stored in the MRAM cell 10_1 is obtained. Next, according to the state transition diagram of the data DATA_1 of FIG. 4A, the desired state corresponding to new data DATA_1 that is to be written into the MRAM cell 10_1, is reached with one or more write operations.

For example, if the current state of the MRAM cell 10_1 is the second state S42 and the desired state of the MRAM cell 10_1 is the fourth state S44, the controller 130 is configured to provide the access current I_w within the current range 12 to the MRAM cell 10_1, so as to change the second state S42 to the third state S43. Next, the controller 130 is configured to provide the access current I_w within the current range 13 to the MRAM cell 10_1, so as to change the

third state S43 to the fourth state S44. Thus, the new data DATA_1 is stored in the MRAM cell 10_1.

FIG. 7 shows an example schematic of an MRAM cell 10_2, in accordance with some embodiments of the disclosure. The MRAM cell 10_2 includes a transistor MM and the MTJ devices 200_3 and 200_4 coupled in series. The transistor MM is coupled between a source line SBL and the MTJ device 200_3, and a gate of the transistor MM is coupled to a word line WL. The MTJ devices 200_3 and 200_4 are stacked between the transistor MM and a bit line BL. The MRAM cell 10_2 is capable of storing two-bit data DATA_2 based on the stacked MTJ devices 200_3 and 200_4. As described above, by applying the access current I_w to the stacked MTJ devices 200_3 and 200_4, the MTJ device 200_3 and/or 200_4 is switched from a parallel magnetic state P_state to an anti-parallel magnetic state AP_state or from the anti-parallel magnetic state AP_state to the parallel magnetic state P_state.

In the MRAM cell 10_2, the free layer 210 of the MTJ device 200_3 is in contact with the free layer 210 of the MTJ device 200_4. In some embodiments, a conducting layer (e.g., an electrode) is formed between the free layers 210 of the MTJ devices 200_3 and 200_4. It should be noted that the MTJ devices 200_3 and 200_4 are different sizes.

In some embodiments, the thickness of the free layers 210 of the MTJ devices 200_3 and 200_4 are the same. Furthermore, the thickness of the barrier layer 220 of the MTJ devices 200_3 and 200_4 are the same, and the thickness of the pinned layer 230 of the MTJ devices 200_3 and 200_4 are the same. The difference between the MTJ devices 200_3 and 200_4 is that the area of the free layer 210, the barrier layer 220 and the pinned layer 230 of the MTJ device 200_3 is greater than that of the MTJ device 200_4. Thus, the equivalent impedances of the MTJ devices 200_3 and 200_4 are different.

In some embodiments, the thickness of the free layers 210 of the MTJ devices 200_3 and 200_4 are the same, and the thickness of the pinned layer 230 of the MTJ devices 200_3 and 200_4 are the same. Furthermore, the area of the free layer 210, the barrier layer 220 and the pinned layer 230 of the MTJ device 200_3 is the same as that of the MTJ device 200_4. The difference between the MTJ devices 200_3 and 200_4 is that the thickness of the barrier layer 220 of the MTJ devices 200_3 and 200_4 are different. Thus, the equivalent impedances of the MTJ devices 200_3 and 200_4 are different.

FIG. 8A shows an example state transition diagram of the data DATA_2 in the MRAM cell 10_2 of FIG. 7, in accordance with some embodiments of the disclosure. FIG. 8B shows an example illustrating the access current I_w corresponding to the state transition diagram of the data DATA_2 in FIG. 8A, in accordance with some embodiments of the disclosure.

In FIG. 8A, the data DATA_2 have four states S51 through S54. In the first state S51 (i.e., “ap”), the MTJ device 200_3 is operated in the anti-parallel magnetic state AP_state and the MTJ device 200_4 is operated in the parallel magnetic state P_state. In the second state S52 (i.e., “aa”), the MTJ devices 200_3 and 200_4 are operated in the anti-parallel magnetic state AP_state. In the third state S53 (i.e., “pa”), the MTJ device 200_3 is operated in the parallel magnetic state P_state and the MTJ device 200_4 is operated in the anti-parallel magnetic state AP_state. In the fourth state S54 (i.e., “pp”), the MTJ devices 200_3 and 200_4 are operated in the parallel magnetic state P_state.

The access current I_w within the current regions II through 14 is a writing current for storing the two-bit data

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DATA_2 in the MRAM cell 10_2. Conversely, the access current I_w within the current region IR (i.e. $-I_{c2} < I_w < I_{c2}$) is a sense current for reading the two-bit data DATA_2 stored in the MRAM cell 10_2.

In some embodiments, the access current I_w within the current region 11 is capable of controlling the magnetization directions of the free layer 210 and the pinned layer 230 of the MTJ device 200_4 as the anti-parallel magnetic state AP_state, and the access current I_w within the current region 13 is capable of controlling the magnetization directions of the free layer 210 and the pinned layer 230 of the MTJ device 200_4 as the parallel magnetic state P_state. The access current I_w within the current region 12 is capable of controlling the magnetization directions of the free layer 210 and the pinned layer 230 of the MTJ device 200_3 as the parallel magnetic state P_state, and the access current I_w within the current region 14 is capable of controlling the magnetization directions of the free layer 210 and the pinned layer 230 of the MTJ device 200_3 as the anti-parallel magnetic state AP_state.

As previously described in FIG. 6, each of the states S51 through S54 has individual equivalent impedance. Thus, by providing the sense signal (current or voltage) to obtain the equivalent impedance of the stacked MTJ devices 200_3 and 200_4 in the MRAM cell 10_2, the current state of the MRAM cell 10_2 is obtained. According to the current state of the MRAM cell 10_2, the data DATA_2 stored in the MRAM cell 10_2 is obtained. Next, according to the state transition diagram of the data DATA_2 of FIG. 8A, the desired state corresponding to new data DATA_2 that is to be written into the MRAM cell 10_2 is reached with one or more write operations.

For example, if the current state of the MRAM cell 10_2 is the fourth state S54 and the desired state of the MRAM cell 10_2 is the second state S52, the controller 130 is configured to provide the access current I_w within the current range 14 to the MRAM cell 10_2, so as to change the fourth state S54 to the first state S51. Next, the controller 130 is configured to provide the access current I_w within the current range 11 to the MRAM cell 10_2, so as to change the first state S51 to the second state S52. Thus, the new data DATA_2 is stored in the MRAM cell 10_2.

FIG. 9 shows an example schematic of an MRAM cell 10_3, in accordance with some embodiments of the disclosure. The MRAM cell 10_3 includes a transistor MM and the MTJ devices 200_5 through 200_8 coupled in series. The transistor MM is coupled between a source line SBL and the MTJ device 200_5, and a gate of the transistor MM is coupled to a word line WL. The MTJ devices 200_5 through 200_8 are stacked between the transistor MM and a bit line BL. The MRAM cell 10_3 is capable of storing three-bit data DATA_3 based on the stacked MTJ devices 200_5 through 200_8. As described above, by applying the access current I_w to the stacked MTJ devices 200_5 through 200_8, the MTJ devices 200_5, 200_6, 200_7 and/or 200_8 is switched from a parallel magnetic state P_state to an anti-parallel magnetic state AP_state or from the anti-parallel magnetic state AP_state to the parallel magnetic state P_state.

In the MRAM cell 10_3, the free layer 210 of the MTJ device 200_5 is in contact with the pinned layer 230 of the MTJ device 200_6, the free layer 210 of the MTJ device 200_6 is in contact with the pinned layer 230 of the MTJ device 200_7, and the free layer 210 of the MTJ device 200_7 is in contact with the pinned layer 230 of the MTJ device 200_8. In some embodiments, a conducting layer (e.g., an electrode) is formed between two adjacent MTJ

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devices. It should be noted that the MTJ devices 200_5 through 200_8 are different sizes.

In some embodiments, the thickness of the free layers 210 of the MTJ devices 200_5 through 200_8 are the same. Furthermore, the thickness of the barrier layer 220 of the MTJ devices 200_5 through 200_8 are the same, and the thickness of the pinned layer 230 of the MTJ devices 200_5 through 200_8 are the same. The difference between the MTJ devices 200_5 through 200_8 is that the area of the free layer 210, the barrier layer 220 and the pinned layer 230 of the MTJ device 200_5 is greater than that of the MTJ device 200_6, and the area of the free layer 210, the barrier layer 220 and the pinned layer 230 of the MTJ device 200_6 is greater than that of the MTJ device 200_7. Furthermore, the area of the free layer 210, the barrier layer 220 and the pinned layer 230 of the MTJ device 200_7 is greater than that of the MTJ device 200_8. Thus, the equivalent impedances of the MTJ devices 200_5 through 200_8 are different.

In some embodiments, the thickness of the free layers 210 of the MTJ devices 200_5 through 200_8 are the same, and the thickness of the pinned layer 230 of the MTJ devices 200_5 through 200_8 are the same. Furthermore, the area of the free layer 210, the barrier layer 220 and the pinned layer 230 of the MTJ devices 200_5 through 200_8 are the same. The difference between the MTJ devices 200_5 through 200_8 is that the thickness of the barrier layer 220 of the MTJ devices 200_5 through 200_8 are different. Thus, the equivalent impedances of the MTJ devices 200_5 through 200_8 are different.

FIG. 10A shows an example state transition diagram of the data DATA_3 in the MRAM cell 10_3 of FIG. 9, in accordance with some embodiments of the disclosure. FIG. 10B shows an example illustrating the access current I_w corresponding to the state transition diagram of the data DATA_3 in FIG. 10A, in accordance with some embodiments of the disclosure. In FIG. 10A, the data DATA_3 have eight states S61 through S68.

In the first state S61 (i.e., "aaaa"), the MTJ devices 200_5 through 200_8 are operated in the anti-parallel magnetic state AP_state.

In the second state S62 (i.e., "aaap"), the MTJ devices 200_5 through 200_7 are operated in the anti-parallel magnetic state AP_state, and the MTJ device 200_8 is operated in the parallel magnetic state.

In the third state S63 (i.e., "aapp"), the MTJ devices 200_5 and 200_6 are operated in the anti-parallel magnetic state AP_state, and the MTJ devices 200_7 and 200_8 are operated in the parallel magnetic state.

In the fourth state S64 (i.e., "app"), the MTJ device 200_5 is operated in the anti-parallel magnetic state AP_state, and the MTJ devices 200_6 through 200_8 are operated in the parallel magnetic state.

In the fifth state S65 (i.e., "pppp"), the MTJ devices 200_5 through 200_8 are operated in the parallel magnetic state.

In the sixth state S66 (i.e., "pppa"), the MTJ devices 200_5 through 200_7 are operated in the parallel magnetic state, and the MTJ device 200_8 is operated in the anti-parallel magnetic state AP_state.

In the sixth state S67 (i.e., "ppaa"), the MTJ devices 200_5 and 200_6 are operated in the parallel magnetic state, and the MTJ devices 200_7 and 200_8 are operated in the anti-parallel magnetic state AP_state.

In the sixth state S68 (i.e., "paaa"), the MTJ device 200_5 is operated in the parallel magnetic state, and the MTJ devices 200_6 through 200_8 are operated in the anti-parallel magnetic state AP_state.

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The access current I_w within the current regions **11** through **18** is a writing current for storing the three-bit data **DATA_3** in the MRAM cell **10_3**. Conversely, the access current I_w within the current region **1R** (i.e. $-I_{c4} < I_w < I_{c4}$) is a sense current for reading the three-bit data **DATA_3** stored in the MRAM cell **10_3**.

The current region **11** is greater than the current value I_{c4} and less than the current value I_{c3} , and the current region **12** is greater than the current value I_{c3} and less than the current value I_{c2} . Furthermore, the current region **13** is greater than the current value I_{c2} and less than the current value I_{c1} , and the current region **14** is greater than the current value I_{c4} . Moreover, the current region **15** is greater than the current value $-I_{c3}$ and less than the current value $-I_{c4}$, and the current region **16** is greater than the current value $-I_{c2}$ and less than the current value $-I_{c3}$. In addition, the current region **17** is greater than the current value $-I_{c1}$ and less than the current value $-I_{c2}$, and the current region **18** is less than the current value $-I_{c1}$.

In some embodiments, the access current I_w within the current region **11** is capable of controlling the magnetization directions of the free layer **210** and the pinned layer **230** of the MTJ device **200_8** as the parallel magnetic state **P_state**, and the access current I_w within the current region **15** is capable of controlling the magnetization directions of the free layer **210** and the pinned layer **230** of the MTJ device **200_8** as the anti-parallel magnetic state **AP_state**. In some embodiments, the access current I_w within the current region **12** is capable of controlling the magnetization directions of the free layer **210** and the pinned layer **230** of the MTJ device **200_7** as the parallel magnetic state **P_state**, and the access current I_w within the current region **16** is capable of controlling the magnetization directions of the free layer **210** and the pinned layer **230** of the MTJ device **200_7** as the anti-parallel magnetic state **AP_state**.

In some embodiments, the access current I_w within the current region **13** is capable of controlling the magnetization directions of the free layer **210** and the pinned layer **230** of the MTJ device **200_6** as the parallel magnetic state **P_state**, and the access current I_w within the current region **17** is capable of controlling the magnetization directions of the free layer **210** and the pinned layer **230** of the MTJ device **200_6** as the anti-parallel magnetic state **AP_state**. In some embodiments, the access current I_w within the current region **14** is capable of controlling the magnetization directions of the free layer **210** and the pinned layer **230** of the MTJ device **200_5** as the parallel magnetic state **P_state**, and the access current I_w within the current region **18** is capable of controlling the magnetization directions of the free layer **210** and the pinned layer **230** of the MTJ device **200_5** as the anti-parallel magnetic state **AP_state**.

FIG. **11** shows an example schematic of an MRAM cell **10_4**, in accordance with some embodiments of the disclosure. The MRAM cell **10_4** includes a transistor **MM** and the MTJ devices **200_9** through **200_x** coupled in series. The transistor **MM** is coupled between a source line **SBL** and the MTJ device **200_9**, and a gate of the transistor **MM** is coupled to a word line **WL**. The MTJ devices **200_9** through **200_x** are stacked between the transistor **MM** and a bit line **BL**. The MRAM cell **10_4** is capable of storing m -bit data **DATA_4** based on the stacked MTJ devices **200_9** through **200_x**. As described above, by applying the access current I_w to the stacked MTJ devices **200_9** through **200_x**, the MTJ devices **200_9** through **200_x** can be switched from a parallel magnetic state **P_state** to an anti-parallel magnetic state **AP_state** or from the anti-parallel magnetic state **AP_state** to the parallel magnetic state **P_state**, respectively.

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In the MRAM cell **10_4**, the free layer **210** of the MTJ device **200_9** is in contact with the pinned layer **230** of the MTJ device **200_10**, and the free layer **210** of the MTJ device **200_10** is in contact with the free layer **210** of the MTJ device **200_11**. In some embodiments, a conducting layer (e.g., an electrode) is formed between two adjacent MTJ devices. Furthermore, the MTJ devices **200_9** through **200_x** are different sizes. For example, the area of the MTJ device **200_9** is greater than that of the MTJ device **200_11**, and the area of the MTJ device **200_11** is greater than that of the MTJ device **200_10**. As described above, the equivalent impedances of the MTJ devices **200_9** through **200_x** are different. Furthermore, the stacked configuration of the MTJ devices **200_9** through **200_x** is determined according to different applications. In other words, the sort order and connection type of the MTJ devices **200_9** through **200_x** are adjustable based on the related applications.

In each MRAM cell, the bit number of data stored in the MRAM cell is determined according to the number of MTJ devices within the MRAM cell. For example, in the MRAM cell **10_4**, the number of MTJ devices **200_9** through **200_x** is N , and the data **DATA_4** have 2^N states. Furthermore, the bit number m of the data **DATA_4** is obtained according to the following equation:

$$m = \log_2(2 \times N).$$

As described above, each of the N states of the data **DATA_4** has individual equivalent impedance. Thus, by providing the sense signal (current or voltage) to obtain the equivalent impedance of the stacked MTJ devices **200_9** through **200_x** in the MRAM cell **10_4**, the current state of the MRAM cell **10_4** is obtained. According to the current state of the MRAM cell **10_4**, the data **DATA_4** stored in the MRAM cell **10_4** is obtained. Next, according to the states of the data **DATA_4**, the desired state corresponding to new data **DATA_4** that is to be written into the MRAM cell **10_4** is reached with one or more write operations.

FIG. **12** shows an example schematic of an MRAM cell **10_5**, in accordance with some embodiments of the disclosure. The MRAM cell **10_5** includes a transistor **MM** and the MTJ devices **200_a** and **200_b** coupled in series. The transistor **MM** is coupled between a source line **SBL** and the MTJ device **200_b**, and a gate of the transistor **MM** is coupled to a word line **WL**. The MTJ devices **200_a** and **200_b** are stacked between the transistor **MM** and a bit line **BL**. The MRAM cell **10_5** is capable of storing data based on the stacked MTJ devices **200_a** and **200_b**. As described above, by applying the access current I_w to the stacked MTJ devices **200_a** and **200_b**, the MTJ device **200_a** and/or **200_b** is switched from a parallel magnetic state **P_state** to an anti-parallel magnetic state **AP_state** or from the anti-parallel magnetic state **AP_state** to the parallel magnetic state **P_state**.

In the MRAM cell **10_5**, the MTJ device **200_b** is formed by the MTJ sub-devices **200_b1** and **200_b2**. The MTJ sub-devices **200_b1** and **200_b2** are coupled in parallel between the conducting layers **270** and **272** (e.g., electrodes). The conducting layers **270** and **272** are formed in different levels. The pinned layers **230** of the MTJ sub-devices **200_b1** and **200_b2** are in contact with the conducting layer **270**, and the free layers **210** of the MTJ sub-devices **200_b1** and **200_b2** are in contact with the conducting layer **272**. Furthermore, the pinned layer **230** of the MTJ device **200_a** is in contact with the conducting layer **272**.

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In some embodiments, the thickness of the free layers **210** of the MTJ device **200_a** and the MTJ sub-devices **200_b1** and **200_b2** are the same. Furthermore, the thickness of the barrier layer **220** of the MTJ device **200_a** and the MTJ sub-devices **200_b1** and **200_b2** are the same, and the thickness of the pinned layer **230** of the MTJ device **200_a** and the MTJ sub-devices **200_b1** and **200_b2** are the same. It should be noted that the MTJ device **200_a** and the MTJ sub-devices **200_b1** and **200_b2** are the same sizes. Specifically, the area of the free layer **210**, the barrier layer **220** and the pinned layer **230** of the MTJ device **200_a** and the MTJ sub-devices **200_b1** and **200_b2** have the same unity areas. The MTJ sub-devices **200_b1** and **200_b2** are coupled in parallel to form the MTJ device **200_b**, and the MTJ device **200_b** has greater area than the MTJ device **200_a**. Therefore, the equivalent impedances of the MTJ devices **200_a** and **200_b** are different. Furthermore, various impedances are provided by coupling multiple unity areas in parallel.

FIG. 13 shows an example schematic of an MRAM cell **10_6**, in accordance with some embodiments of the disclosure. The MRAM cell **10_6** includes a transistor MM and the MTJ devices **200_c** and **200_d** coupled in series. The transistor MM is coupled between a source line SBL and the MTJ device **200_d**, and a gate of the transistor MM is coupled to a word line WL. The MTJ devices **200_c** and **200_d** are stacked between the transistor MM and a bit line BL. The MRAM cell **10_6** is capable of storing data based on the stacked MTJ devices **200_c** and **200_d**. As described above, by applying the access current I_w to the stacked MTJ devices **200_c** and **200_d**, the MTJ device **200_c** and/or **200_d** is switched from a parallel magnetic state P_state to an anti-parallel magnetic state AP_state or from the anti-parallel magnetic state AP_state to the parallel magnetic state P_state.

In the MRAM cell **10_6**, the MTJ device **200_d** is formed by the MTJ sub-devices **200_d1** and **200_d2**. The MTJ sub-devices **200_d1** and **200_d2** are coupled in parallel between the conducting layer **270** (e.g., electrode) and the pinned layer **230** of the MTJ device **200_c**. The pinned layers **230** of the MTJ sub-devices **200_d1** and **200_d2** are in contact with the conducting layer **270**, and the free layers **210** of the MTJ sub-devices **200_d1** and **200_d2** are in contact with the pinned layers **230** of the MTJ devices **200_c**.

In some embodiments, the thickness of the free layers **210** of the MTJ device **200_c** and the MTJ sub-devices **200_d1** and **200_d2** are the same. Furthermore, the thickness of the barrier layer **220** of the MTJ device **200_c** and the MTJ sub-devices **200_d1** and **200_d2** are the same, and the thickness of the pinned layer **230** of the MTJ device **200_c** and the MTJ sub-devices **200_d1** and **200_d2** are the same. Furthermore, the area of the free layer **210** and the barrier layer **220** of the MTJ device **200_c** and the MTJ sub-devices **200_d1** and **200_d2** are the same. The area of the pinned layer **230** of the MTJ sub-devices **200_d1** and **200_d2** are the same. In FIG. 13, the area of the pinned layer **230** of the MTJ device **200_c** is greater than that of the MTJ sub-devices **200_d1** and **200_d2**. In some embodiments, the area of the pinned layer **230** of the MTJ device **200_c** is equal to that of the conducting layer **270**.

As described above, the MTJ sub-devices **200_d1** and **200_d2** are coupled in parallel to form the MTJ device **200_d**, and the MTJ device **200_d** has greater area than the MTJ device **200_c**. Therefore, the equivalent impedances of the MTJ devices **200_c** and **200_d** are different.

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FIG. 14 shows an example schematic of an MRAM cell **10_6**, in accordance with some embodiments of the disclosure. The MRAM cell **10_6** includes a transistor MM and the MTJ devices **200_e** and **200_f** coupled in series. The transistor MM is coupled between a source line SBL and the MTJ device **200_e**, and a gate of the transistor MM is coupled to a word line WL. The MTJ device **200_e** is coupled between the transistor MM and the conducting layer **272**, and the MTJ device **200_f** is coupled between the conducting layer **272** and the bit line BL. The MRAM cell **10_7** is capable of storing data based on the MTJ devices **200_e** and **200_f**. As described above, by applying the access current I_w to the MTJ devices **200_e** and **200_f**, the MTJ device **200_e** and/or **200_f** is switched from a parallel magnetic state P_state to an anti-parallel magnetic state AP_state or from the anti-parallel magnetic state AP_state to the parallel magnetic state P_state.

In the MRAM cell **10_7**, the pinned layers **230** of the MTJ device **200_e** is in contact with the conducting layer **270_1**, and the free layers **210** of the MTJ device **200_e** is in contact with the conducting layer **272**. Furthermore, the pinned layers **230** of the MTJ device **200_f** is in contact with the conducting layer **270_2**, and the free layers **210** of the MTJ device **200_f** is in contact with the conducting layer **272**. The conducting layers **270_1** and **270_2** are formed in the same level that is different from the conducting layer **272**. It should be noted that the MTJ devices **200_e** and **200_f** are different sizes.

In some embodiments, the thickness of the free layers **210** of the MTJ devices **200_e** and **200_f** are the same. Furthermore, the thickness of the barrier layer **220** of the MTJ devices **200_e** and **200_f** are the same, and the thickness of the pinned layer **230** of the MTJ devices **200_e** and **200_f** are the same. The difference between the MTJ devices **200_e** and **200_f** is that the area of the free layer **210**, the barrier layer **220** and the pinned layer **230** of the MTJ device **200_e** is greater than that of the MTJ device **200_f**. Thus, the equivalent impedances of the MTJ devices **200_e** and **200_f** are different.

In some embodiments, the thickness of the free layers **210** of the MTJ devices **200_e** and **200_f** are the same, and the thickness of the pinned layer **230** of the MTJ devices **200_e** and **200_f** are the same. Furthermore, the area of the free layer **210**, the barrier layer **220** and the pinned layer **230** of the MTJ device **200_e** is the same as that of the MTJ device **200_f**. The difference between the MTJ devices **200_e** and **200_f** is that the thickness of the barrier layer **220** of the MTJ devices **200_e** and **200_f** are different. Thus, the equivalent impedances of the MTJ devices **200_e** and **200_f** are different.

In some embodiments, the MTJ devices **200_e** and **200_f** are formed by multiple MTJ sub-devices coupled in parallel or in serial, i.e., the MTJ devices **200_e** and **200_f** are not single MTJ device. In the MRAM cell **10_7**, the MTJ devices **200_e** and **200_f** are formed in the same level. In some embodiments, multiple MTJ devices are coupled between the MTJ devices **200_e** and **200_f**, and the MTJ devices and the MTJ devices **200_e** and **200_f** are formed in the same level. A connecting line **281** coupled between the MTJ device **200_e** and the transistor MM is formed below the MTJ device **200_e**. Similarly, a connecting line **282** coupled between the MTJ device **200_f** and the bit line BL is formed below the MTJ device **200_e**. In some embodiments, the connecting lines **281** and **282** are formed in the same metal layer under the MTJ devices **200_e** and **200_f**. In some embodiments, the connecting lines **281** and **282** are formed in the different metal layers under the MTJ devices

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200_e and 200_f. In FIG. 14, no connecting line is present over the MTJ devices 200_e and 200_f, and the routing metal layers over the MTJ devices 200_e and 200_f can be omitted.

Embodiments of MRAM including multiple MTJ devices are provided. In the MRAM cell, the MTJ devices are coupled in series and stacked vertically, and the stacked MTJ devices does not increase the area of the MRAM cell. Furthermore, the MTJ devices are different sizes, and the equivalent impedances of the MTJ devices are different. According to the state transition diagram of the stacked MTJ devices, the access current I_w is used to read the data stored in the MRAM cell or write the data into the MRAM cell 10_1 with one or more write operations. Compared with the conventional MRAM cell including a transistor and a MTJ device, the MRAM cell including the stacked MTJ devices is capable of storing more bits without increasing area.

In some embodiments, a magnetic random access memory (MRAM) cell is provided. The MRAM cell includes a plurality of stacked magnetic tunnel junction (MTJ) devices coupled in serial and coupled between a bit line and a source line. The stacked MTJ devices have different sizes. Each of the stacked MTJ devices includes a free layer, a pinned layer and a barrier layer between the free layer and the pinned layer. The free layers of two adjacent stacked MTJ devices are in direct contact with each other.

In some embodiments, a magnetic random access memory (MRAM) cell is provided. The MRAM cell includes a plurality of stacked magnetic tunnel junction (MTJ) devices coupled in serial and coupled between a bit line and a source line. Each of the stacked MTJ devices includes a free layer, a pinned layer and a barrier layer between the free layer and the pinned layer. A first stacked MTJ device of the stacked MTJ devices is coupled to the source line. The free layers of two adjacent stacked MTJ devices between the first stacked MTJ device and the bit line are in direct contact with each other. Size of the first stacked MTJ device is greater than that of the two adjacent stacked MTJ devices.

In some embodiments, a magnetic random access memory (MRAM) is provided.

The MRAM includes a plurality of memory cells arranged in a memory array. Each of the memory cells includes a plurality of magnetic tunnel junction (MTJ) devices coupled in series for storing multiple-bit data and a transistor. In each of the memory cells, one of the MTJ devices is coupled to the transistor, and the MTJ devices have different sizes. In each of the memory cells, each of the MTJ devices includes a free layer, a pinned layer and a barrier layer between the free layer and the pinned layer and the pinned layer of the one of the MTJ devices is directly coupled to the transistor. In each of the memory cells, the free layers of two adjacent MTJ devices are in direct contact with each other.

The foregoing outlines nodes of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

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What is claimed is:

1. A magnetic random access memory (MRAM) cell, comprising:

a plurality of stacked magnetic tunnel junction (MTJ) devices coupled in serial and coupled between a bit line and a source line;

wherein the stacked MTJ devices have different sizes, wherein each of the stacked MTJ devices comprises a free layer, a pinned layer and a barrier layer between the free layer and the pinned layer,

wherein the free layers of two adjacent stacked MTJ devices are in direct contact with each other.

2. The MRAM cell as claimed in claim 1, wherein the number of the stacked MTJ devices is N, and a bit number of data stored in the MRAM cell is $\text{Log } 2(2N)$.

3. The MRAM cell as claimed in claim 1, wherein thickness of the pinned layers of the stacked MTJ devices are the same, and thickness of the free layers of the stacked MTJ devices are the same.

4. The MRAM cell as claimed in claim 1, wherein the free layers of the two adjacent stacked MTJ devices have different areas.

5. The MRAM cell as claimed in claim 4, wherein thickness of the barrier layers of the two adjacent stacked MTJ devices are the same.

6. The MRAM cell as claimed in claim 1, wherein the free layers of the two adjacent stacked MTJ devices have the same areas.

7. The MRAM cell as claimed in claim 6, wherein thicknesses of the barrier layers of the two adjacent stacked MTJ devices are different.

8. A magnetic random access memory (MRAM) cell, comprising:

a plurality of stacked magnetic tunnel junction (MTJ) devices coupled in serial and coupled between a bit line and a source line;

wherein each of the stacked MTJ devices comprises a free layer, a pinned layer and a barrier layer between the free layer and the pinned layer,

wherein a first stacked MTJ device of the stacked MTJ devices is coupled to the source line,

wherein the free layers of two adjacent stacked MTJ devices between the first stacked MTJ device and the bit line are in direct contact with each other,

wherein size of the first stacked MTJ device is greater than that of the two adjacent stacked MTJ devices.

9. The MRAM cell as claimed in claim 8, wherein the number of the stacked MTJ devices is N, and a bit number of data stored in the MRAM cell is $\text{Log } 2(2N)$.

10. The MRAM cell as claimed in claim 8, wherein the stacked MTJ devices have different sizes.

11. The MRAM cell as claimed in claim 8, wherein area of the first stacked MTJ device is greater than that of the two adjacent stacked MTJ devices.

12. The MRAM cell as claimed in claim 8, wherein the two adjacent stacked MTJ devices comprises a second stacked MTJ device close to the source line and a third stacked MTJ device away from the source line, wherein area of the second stacked MTJ device is less than that of the third stacked MTJ device.

13. The MRAM cell as claimed in claim 12, wherein area of the third stacked MTJ device is less than that of the first stacked MTJ device.

14. A magnetic random access memory (MRAM), comprising:

a plurality of memory cells arranged in a memory array, wherein each of the memory cells comprises a plurality

of magnetic tunnel junction (MTJ) devices coupled in series for storing multiple-bit data and a transistor, wherein in each of the memory cells, one of the MTJ devices is coupled to the transistor, and the MTJ devices have different sizes, 5

wherein in each of the memory cells, each of the MTJ devices comprises a free layer, a pinned layer and a barrier layer between the free layer and the pinned layer, and the pinned layer of the one of the MTJ devices is directly coupled to the transistor, 10

wherein in each of the memory cells, the free layers of two adjacent MTJ devices are in direct contact with each other.

15. The MRAM as claimed in claim **14**, wherein the MTJ devices have different areas in each of the memory cells. 15

16. The MRAM as claimed in claim **14**, wherein in each of the memory cells, the number of the MTJ devices is N, and a bit number of data stored in the MRAM cell is $\log_2(2N)$.

17. The MRAM as claimed in claim **14**, wherein in each of the memory cells, the free layers of the two adjacent MTJ devices have different areas. 20

18. The MRAM as claimed in claim **17**, wherein thickness of the barrier layers of the two adjacent MTJ devices are the same. 25

19. The MRAM as claimed in claim **14**, wherein in each of the memory cells, the free layers of the two adjacent MTJ devices have the same areas.

20. The MRAM as claimed in claim **19**, wherein thicknesses of the barrier layers of the two adjacent MTJ devices are different. 30

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